

STRESSDREAM: Steering Video World Models for Robust Policy Evaluation and Improvement

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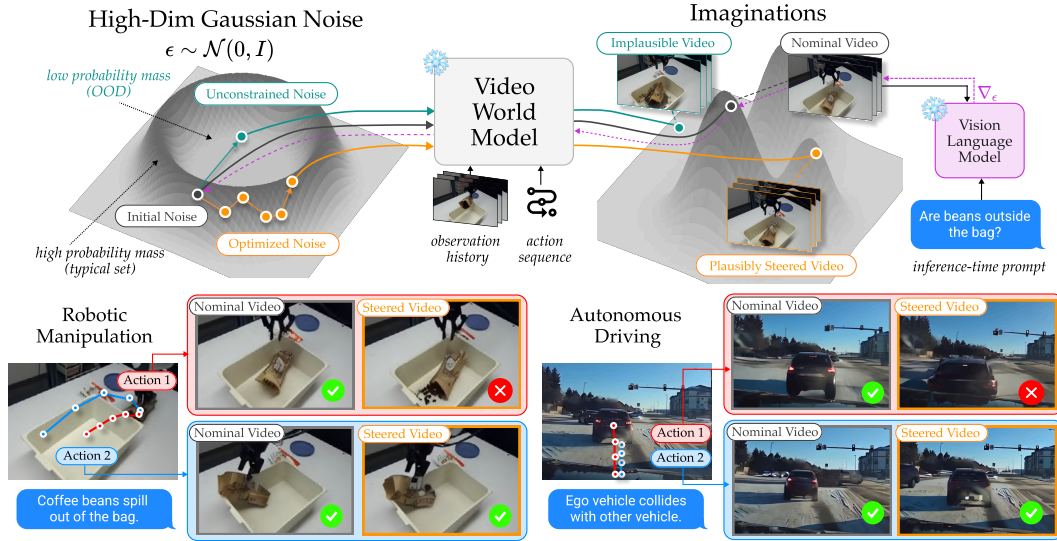


Figure 1: Overview of STRESSDREAM. STRESSDREAM steers video world model generation into plausible, critical imaginations by optimizing the initial noise of the diffusion model. *Top:* The optimization is guided by VLMs, while preventing the high-dimensional noise from drifting into OOD, preserving plausibility. *Bottom:* STRESSDREAM enables robust policy evaluation and improvement through steered imaginations.

Abstract: Video world models promise scalable policy evaluation and improvement by predicting realistic future observations conditioned on robot actions. While these models can capture a distribution over possible futures, policy evaluation and improvement typically rely on a single nominal imagination that may miss critical yet plausible outcomes. To address this, we propose STRESSDREAM, an algorithm for steering video world-model generations at inference-time toward critical outcomes within the distribution of plausible futures. Our key idea is to optimize the diffusion model’s initial noise using two complementary objectives: a semantic objective from a Vision-Language Model that reasons about generated videos, and a distributional objective that prevents the high-dimensional noise from drifting away from the Gaussian prior, which can otherwise lead to implausible generations. With state-of-the-art video world models for autonomous driving and robotic manipulation, we show that STRESSDREAM can effectively steer generation toward semantically meaningful events, such as safety-critical events or task failures, while preserving plausibility. By considering the distribution of outcomes, STRESSDREAM enables robust policy evaluation by identifying actions that risk failure, and improves policies by promoting robust actions that avoid failures even under adverse but plausible futures. Video results can be found at [Project Website](#).

Keywords: World Models, Inference-Time Steering, Policy Learning and Evaluation

1 Introduction

Video world models (WMs), or learned simulators of physical environments, have attracted significant interest in robotics, promising scalable policy evaluation and improvement with substantially less costly real-world interaction [1]. At their core, video WMs are generative models, such as diffusion and flow matching [2, 3, 4], learning *distributions over future observations* conditioned on robot actions, and thereby capturing the inherent uncertainty of physical interaction [5, 6, 7, 8] (*top*, Fig. 1). For example, in robotic manipulation, dropping an open bag from high above the table may or may not cause its contents to spill, leading to either task success or failure; in autonomous driving, the same ego-vehicle decision can elicit qualitatively different safety-critical outcomes depending on the responses of surrounding agents, such as braking or maintaining speed.

Despite learning rich distributions over future outcomes, WMs today are predominantly used by generating a *single, nominal* imagination during policy evaluation or improvement, discarding the outcome uncertainty they are trained to model [6, 9]. Naively sampling multiple imaginations from WMs is not only computationally prohibitive (e.g., minutes per sample [10, 11]), but more fundamentally ill-suited to the role of WMs: exploring plausible, high-impact outcomes of robot actions for *robust* policy evaluation and improvement. For example, when a manipulator drops an open bag from high above the table, the WM’s distribution may include outcomes both with and without spilling; in contrast, when the bag is placed close to the table, spilling may be unlikely or absent from the outcome distribution (*bottom*, Fig. 1). The ability to imagine critical outcomes can therefore allow the latter action to be preferred during policy evaluation, while the former is downweighted or suppressed during policy improvement. This motivates the central question of our work:

How can we steer video world models at inference time to generate plausible, high-impact outcomes for robust policy evaluation and improvement?

Our key idea is to optimize the diffusion model’s *initial noise* at inference time, steering WM generations toward semantically meaningful events (e.g., spills or collisions), while keeping them plausible under the learned outcome distribution. The challenge, however, is to guide this optimization meaningfully and reliably in *extremely high-dimensional* noise spaces (e.g., 1M dims [10]): the optimization must score generated videos by reasoning about nuanced, scene-dependent target events, while naive optimization can easily push the initial noise away into out-of-distribution (OOD) regions [12, 13], producing implausible videos. To address this, we combine a *semantic* objective, provided by a Vision-Language Model (VLM) that reasons about *what happens in the generated video*, with a *distributional* objective that preserves the statistical properties of the standard Gaussian noise prior. Together, these objectives provide dense gradients for searching the high-dimensional noise space while keeping the optimized noise within the prior’s in-distribution, enabling video WMs to explore plausible, high-impact outcome distributions for robust policy evaluation and improvement.

Statement of Contributions. We formalize the first inference-time steering method for video world models, and provide a practical instantiation based on noise optimization for robotic control. We validate STRESSDREAM on three video world models: state-of-the-art models for autonomous driving [10] and robotic manipulation [11], as well as a simulated navigation task for controlled analysis. We first show, in a setting with access to the true system dynamics, that STRESSDREAM enables robust evaluation of actions by steering imaginations toward pessimistic outcomes only when they are plausible. We then show that STRESSDREAM scales to real-world video world models [10, 11], enabling robust policy evaluation by detecting potential failures with substantially higher recall (54% $\rightarrow$ 94%). Finally, we show that this robust evaluation leads to policy improvement by promoting actions that avoid potential failures, improving the success rate of a Vision-Language-Action (VLA) policy [14] from 39% to 71%.

2 Related Works

Video World Models for Robotics. Video world models predict future observations conditioned on actions, enabling learned simulation for policy evaluation and improvement [15, 16, 17, 18, 19,

20, 21, 22, 23]. Recent video generation models have further improved the realism and temporal consistency of these predictions in real-world robotic settings [24, 10, 25, 11]. Under partial observability, future predictions are naturally uncertain and multimodal [26, 27, 8], making diffusion- and flow-matching-based world models well suited for modeling rich distributions over future trajectories [5, 6, 7, 8]. However, many existing methods rely on a single nominal rollout [6, 19, 11, 28], limiting their ability to reason over outcome distributions for downstream tasks.

Inference-time Steering of Generative Models. Inference-time steering guides a generative model toward samples that maximize an inference-time reward [13, 29, 30, 31]. One common approach is noise optimization, which steers generation by optimizing the noise of the denoising process [32, 33, 34, 35, 13], which has shown promise in computer vision, including improving image quality [33, 36, 37], approximating diffusion guidance [38], and generating rare or diverse concepts [12, 39, 40, 41], and more recently in robotics, for adapting diffusion policies [42]. However, noise optimization has not yet been explored for video world models [10, 11, 22, 1], where the high-dimensional noise space poses a central challenge: generations can become implausible once the noise deviates from the prior [12, 13, 35]. Moreover, because this often requires backpropagating through the denoising process, prior work often relies on step-distilled models to make optimization tractable [43, 35, 37], while recent methods use score distillation to approximate gradients [38].

3 Background: Action-conditioned Video World Models

Video World Models. An action-conditioned video world model f_θ predicts future observations $\mathbf{o} := o_{t+1:t+H}$, where each observation $o = [I^1, \dots, I^n, q] \in \mathcal{O}$ consists of n camera images I^i and proprioceptive input q , such as the robot joint configuration. The prediction is made over horizon H , conditioned on an observation history $\mathbf{o}_{\text{hist}} := o_{t-H:t}$ and a future action sequence $\mathbf{a} := a_{t:t+H}$:

$$\mathbf{o} \sim f_\theta(\cdot \mid \mathbf{o}_{\text{hist}}, \mathbf{a}), \quad \mathbf{o}_{\text{hist}} := o_{t-H:t}, \quad \mathbf{a} := a_{t:t+H}. \quad (1)$$

Diffusion Models as Video World Models. Recent video world models often instantiate f_θ as a diffusion model [10, 44, 45, 11, 46], learning a continuous transformation between the data distribution, $\mathbf{x}^0 := \mathbf{o} \sim p_{\text{data}}(\mathbf{o})$, and a tractable noise distribution, typically a standard Gaussian $\mathbf{x}^T := \epsilon \sim \mathcal{N}(0, \mathbf{I})$. The generative process is defined through a probability path between the initial noise distribution ($\mathbf{x}^T$) and the target distribution ($\mathbf{x}^0$) indexed by diffusion time $\tau \in [0, T]$,

$$\mathbf{x}^\tau = \alpha^\tau \mathbf{x} + \sigma^\tau \epsilon, \quad \text{where} \quad \mathbf{x}^0 := \mathbf{o}, \quad \mathbf{x}^T := \epsilon \sim \mathcal{N}(0, \mathbf{I}), \quad (2)$$

where α^τ is a decreasing function of τ and σ^τ is an increasing function. Generation proceeds in the reverse direction by iteratively mapping a noisier sample to one at a less noisy diffusion time:

$$d^\tau(\mathbf{x}^\tau, \mathbf{o}_{\text{hist}}, \mathbf{a}) := \mathbf{x}^{\tau-1} = \frac{\alpha^{\tau-1}}{\alpha^\tau} (\mathbf{x}^\tau - \sigma^\tau g_\theta^\tau(\mathbf{x}^\tau, \mathbf{o}_{\text{hist}}, \mathbf{a})) + \sigma^{\tau-1} g_\theta^{\tau-1}(\mathbf{x}^\tau, \mathbf{o}_{\text{hist}}, \mathbf{a}), \quad (3)$$

where g_θ^τ denotes the network output (see Appendix A). Starting from an initial Gaussian sample $\mathbf{x}^T = \epsilon$, repeated application of this reverse update produces the predicted observation $\mathbf{x}^0 = \mathbf{o}$.

Initial Noise as a Control Variable. Following (3), for fixed conditioning inputs $(\mathbf{o}_{\text{hist}}, \mathbf{a})$, the imaginations $\mathbf{o}$ are a *deterministic* function of the initial noise: $\mathbf{o} = f_\theta(\epsilon, \mathbf{o}_{\text{hist}}, \mathbf{a})$. This implies that selecting the initial noise effectively controls which sample is generated from the world model.

4 Problem Setup: Steering Video World Models for Robust Control

Robust Control with Video World Models. Given an action-conditioned video world model f_θ , we aim to find an optimal action sequence $\mathbf{a}^*$ by evaluating the imagined future outcomes of candidate action sequences, which can then be used to improve robotic policies. The predicted future observations $\mathbf{o}$ are scored by a test-time criterion function $C_\psi^{\text{test}}(\mathbf{o}) \in \mathbb{R}$, parameterized by ψ , which measures whether task-relevant failure events occur in the generated videos, such as spills or collisions. These scores can be used to select actions that minimize failure risk online, as in model-predictive control [19, 47], or to train a policy $\pi(\mathbf{a} \mid \mathbf{o}_{\text{hist}})$ using world-model rollouts [11, 9].

114 However, when multiple futures are possible under the same action sequence, *robust* control [48, 49]
 115 should account for the distribution of outcomes that are plausible under the learned world model:

Inner opt.: high-dimensional noise (e.g., $\sim 1\text{M}$ dims) generating the worst-case plausible future

$$\mathbf{a}^* = \underset{\mathbf{a} \in \mathcal{A}}{\operatorname{argmin}} \underset{\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})}{\operatorname{argmax}} \mathcal{C}_{\psi}^{\text{test}}(f_{\theta}(\boldsymbol{\epsilon}, \mathbf{o}_{\text{hist}}, \mathbf{a})), \quad (4)$$

Outer opt.: action robust to the distribution over plausible outcomes

116 where the inner optimization steers the WM generation toward a worst-case plausible future that
 117 maximizes the criterion score, while the outer optimization selects a robust action sequence from
 118 the admissible set $\mathcal{A}$ that keeps the failure score low across the distribution of plausible futures,
 119 even under such worst-case outcomes. $\mathcal{A}$ may be a discrete set in a sampling-based solver, or a
 120 continuous action space over which a policy is optimized. Here, solving the inner optimization is
 121 challenging because it requires optimization over a high-dimensional initial noise space.

123 **Steering Generation via Inference-time Noise Optimization.** Since the initial noise acts as a
 124 control variable for the generation of WMs (Sec. 3), the goal of the inner optimization corresponds
 125 to steering the denoising process toward a worst-case generation that remains plausible under the
 126 learned model. However, because evaluating each noise requires an expensive denoising process,
 127 repeated random sampling is highly inefficient and likely to miss rare but plausible critical outcomes
 128 within a limited sampling budget. Instead, we leverage the *gradient* of a differentiable criterion
 129 function to steer generation more efficiently, optimizing the initial noise directly via gradient ascent:

$$\boldsymbol{\epsilon}_{i+1} = \boldsymbol{\epsilon}_i + \eta \nabla_{\boldsymbol{\epsilon}_i} [\mathcal{C}_{\psi}^{\text{test}}(f_{\theta}(\boldsymbol{\epsilon}_i, \mathbf{o}_{\text{hist}}, \mathbf{a}))], \quad (5)$$

130 where η denotes a step size. Although this first-order gradient ascent requires a small number of
 131 iterative update steps, each involving forward and backward passes through both the world model
 132 and the criterion function, the dense gradient signal makes it substantially more efficient than zeroth-
 133 order random sampling for discovering initial noises that induce adverse video generations.

134 **Challenge: High-Dimensional Noise Space.** While gradient-based optimization in (5) can steer
 135 diffusion model generation at inference time [35, 13], applying it to video world models for robotic
 136 control is challenging because their noise spaces are extremely high-dimensional, e.g., $H \times C \approx 1\text{M}$
 137 in driving [10] and $\approx 50\text{K}$ in manipulation [11]. The optimization must guide generation toward
 138 meaningful, scene-dependent target events while preserving plausibility under the learned WM dis-
 139 tribution. However, naive optimization can easily push the noise into out-of-distribution (OOD) re-
 140 gions, producing implausible videos (Fig. 1), and the test-time criterion $\mathcal{C}_{\psi}^{\text{test}}$ must be differentiable
 141 while reasoning about nuanced, task-relevant events that vary across scenes. Moreover, solving (5) is
 142 inefficient because exact gradients require backpropagating through the iterative denoising process.

143 5 Noise Optimization in High-Dimensional Video World Models

144 To address the challenges of optimizing high-dimensional noise, we propose STRESSDREAM. We
 145 define the optimization objective $\mathcal{C}_{\psi}^{\text{test}}$ by combining a *semantic* objective using Vision-Language
 146 Models (VLMs), which use inference-time text specifications of target events to reason about
 147 whether the target event occurs in the generated video (Sec. 5.1), with a *distributional* objec-
 148 tive (Sec. 5.2) that preserves the statistical properties of the standard Gaussian noise prior, thereby
 149 maintaining the plausibility of WM generations. Moreover, we approximate the noise gradient
 150 to avoid full backpropagation through the iterative denoising process, enabling efficient optimiza-
 151 tion (Sec. 5.3).

152 5.1 Steering into Inference-time Target Events with Vision-Language-Model Gradients

153 Since video world models operate across diverse scenes and tasks, the steering target may change
 154 with the policy context, requiring a *semantic* criterion that can score nuanced, scene-dependent def-
 155 initions of critical events. We therefore leverage the general video-understanding capability of a

Vision-Language Model (VLM) as the semantic objective. Given an inference-time text specification of the target event $\mathbf{l}$, the VLM measures how well the generated video $\mathbf{o}$ matches the prompt. For example, in robotic manipulation, one may specify “the coffee beans spill,” while in autonomous driving, one may specify “a collision with the front car occurs.” To obtain a differentiable score, we use Qwen-VL [50] and prompt it to output a single token, yes or no, indicating whether $\mathbf{o}$ matches $\mathbf{l}$. We then define the criterion as the difference in log token (See Appendix. B.3 for details):

$$\mathcal{C}_{\psi}^{\text{VLM}}(\mathbf{o}; \mathbf{l}) = \log p_{\text{VLM}}(\text{yes} | \mathbf{o}, \mathbf{l}) - \log p_{\text{VLM}}(\text{no} | \mathbf{o}, \mathbf{l}). \quad (6)$$

5.2 Preserving Generation Plausibility by Keeping Noise in the Typical Set

Diffusion models are trained to generate samples from noise drawn from a Gaussian prior. When the noise leaves the *typical set*, where most training-time noise samples lie, the resulting out-of-distribution (OOD) noise can produce implausible videos that are unlikely under the WM distribution or have degraded visual quality. Importantly, in high dimensions, the typical set is not the same as the region of highest probability density: some noise vectors, such as the zero vector, may have high density but are extremely unlikely to be sampled from the Gaussian prior (See Appendix B.1 for more details.). Since gradient-based noise optimization can perturb the noise away from this typical set [12, 39, 35, 13], inducing atypical coordinate patterns or structured spectral artifacts, we employ a *distributional objective* $\mathcal{C}^{\text{dist}}(\epsilon)$ that encourages the noise to remain within the *typical set* of the Gaussian prior [51, 52], with a combination of functions of the noise that capture key statistics of high-dimensional Gaussian noise: $\mathcal{C}^{\text{dist}}(\epsilon) = \lambda_1 \mathcal{C}_{\text{norm}}(\epsilon) + \lambda_2 \mathcal{C}_{\text{iso}}(\epsilon) + \lambda_3 \mathcal{C}_{\text{spec}}(\epsilon)$.

Norm Concentration. First, since the squared norm of Gaussian noise follows a chi-square distribution, $\|\epsilon\|_2^2 \sim \chi_D^2$, it concentrates sharply around the noise dimension D [12]. We therefore penalize deviations of the noise radius from the typical Gaussian shell: $\mathcal{C}_{\text{norm}}(\epsilon) := (\|\epsilon\|_2^2 - D)^2$.

Isotropy. While norm concentration enforces typicality at a *global* scale, the optimized noise may still contain *local* correlations or structured coordinate patterns that are unlikely under an i.i.d. Gaussian prior [13]. To encourage coordinate-wise independence, we randomly permute the entries of ϵ and partition them into m subvectors $\{\epsilon_i\}_{i=1}^m \subset \mathbb{R}^k$, where $D = mk$. Under the Gaussian prior, these subvectors should behave as i.i.d. samples from $\mathcal{N}(\mathbf{0}, \mathbf{I}_k)$, and therefore their empirical second moment $\hat{\Sigma} = \frac{1}{m} \sum_{i=1}^m \epsilon_i \epsilon_i^T$ should be close to $\mathbf{I}_k$. We therefore penalize deviations from blockwise isotropy as $\mathcal{C}_{\text{iso}}(\epsilon) := \frac{1}{k} \|\hat{\Sigma} - \mathbf{I}_k\|_F^2$, averaged over multiple random permutations.

Spectral Whiteness. Even when the noise remains typical in coordinate space, gradient-based optimization can introduce structured *frequency-domain* artifacts [53]. Since standard Gaussian noise has a flat expected power spectrum [54, 55], we encourage the optimized noise to remain spectrally white. Let $\mathcal{F}$ denote a unitary discrete Fourier transform, and compute the power spectrum $\mathbf{P} = |\mathcal{F}(\epsilon)|^2$. We aggregate $\mathbf{P}$ into B frequency bins to obtain bin-averaged powers $\{\hat{p}_b\}_{b=1}^B$. To preserve spectral flatness, we minimize the variance of these bin-averaged powers, $\mathcal{C}_{\text{spec}}(\epsilon) := \frac{1}{B} \sum_{b=1}^B (\hat{p}_b - \bar{p})^2$, where $\bar{p} = \frac{1}{B} \sum_{b=1}^B \hat{p}_b$ is the mean power across bins.

5.3 Approximating the Noise Gradients for Efficient Steering of Video World Models

Noise optimization in (5) requires computing the gradient of the criterion with respect to the initial noise, $\nabla_{\epsilon} \mathcal{C}_{\psi}(f_{\theta}(\epsilon, \mathbf{o}_{\text{hist}}, \mathbf{a})) = \nabla_{\epsilon} \mathcal{C}_{\psi}(d^1(\cdots d^T(\epsilon, \mathbf{o}_{\text{hist}}, \mathbf{a})))$, where $d^T(\cdot)$ denotes one denoising step in (3). Although diffusion models are differentiable in principle, backpropagating through the full iterative denoising process (e.g., 50 steps [10, 11]) can incur high memory cost and vanishing gradient [35, 37, 38]. To make optimization practical, we adopt a score-distillation-style approximation [38], which approximates the gradient with respect to the initial noise using the gradient at the generated sample: $\nabla_{\epsilon} \mathcal{C}_{\psi}(f_{\theta}(\epsilon, \mathbf{o}_{\text{hist}}, \mathbf{a})) \approx \beta \nabla_{\mathbf{o}} \mathcal{C}_{\psi}(\mathbf{o})$, where β is a scalar scaling coefficient. This approximation only requires backpropagation through the differentiable criterion function, avoiding full backpropagation through the iterative denoising process (See Appendix. B.4 for details.)

In sum, we perform gradient-based optimization as in (5) to steer the WM generations in (4), but with three key components: semantic objective, distributional objective, and gradient approximation:

$$\epsilon_{i+1} = \epsilon_i + \eta \left[\beta \nabla_{f_\theta}(\epsilon_i, \mathbf{o}_{\text{hist}}, \mathbf{a}) \quad \mathcal{C}_\psi^{\text{VLM}}(f_\theta(\epsilon_i, \mathbf{o}_{\text{hist}}, \mathbf{a}); \mathbf{l}) - \nabla_{\epsilon_i} \mathcal{C}^{\text{dist}}(\epsilon_i) \right]. \quad (7)$$

↑ Grad. Approx.
↑ Semantic (VLM)
↑ Distributional (Typical Set)

6 How Accurately Can We Steer Video World Models?

We first study STRESSDREAM in a controlled setting where the ground-truth system dynamics are known. This allows us to directly assess whether steering can explore a pretrained video world model to identify potential critical futures (i.e., failures), while preserving the plausibility of the generation.

Naughty 3D Dubins Car. We consider a discrete-time 3D Dubins car with state $s = [p_x, p_y, \theta]$, continuous angular-velocity action $a_t \in [-1.25, 1.25]$ rad/s, fixed speed $v = 1$ m/s, and timestep $\Delta t = 0.05$ s. The safety score is defined as $\mathcal{C}(s) = p_x^2 + p_y^2 - 0.25^2$, inducing a circular failure set of radius 0.25 m centered at the origin, $\mathcal{F} := \{s \mid \mathcal{C}(s) \leq 0\}$. The vehicle randomly flips the sign of the control input, where $\delta_t = -1$ with probability $p = 0.2$ and $\delta_t = 1$ otherwise:

$$s_{t+1} = s_t + \Delta t [v \cos(\theta_t), v \sin(\theta_t), \delta_t a_t], \quad \delta_t \in \{-1, 1\}. \quad (8)$$

Video World Model. We train an action-conditioned world model with an offline dataset of $N_{\text{train}} = 4,000$ observation-action trajectories annotated with ground-truth safety scores, where each observation is a $3 \times 128 \times 128$ RGB image of the environment and vehicle (See Fig. 2c), and actions are randomly sampled. We train a safety score predictor on observations with the ground-truth scores, and train a diffusion model, $f_\theta(o_{t+1} \mid \epsilon, o_t, a_t)$, to generate one-step futures ($H = 1$) conditioned on the current observation and action, with a noise dimension of 1024.

Evaluation Metrics. We construct an evaluation dataset of 5,000 random trajectories. For each trajectory, we compute the minimum safety score achievable under (8), which determines whether the trajectory has a risk of entering the failure set. We then perform autoregressive world-model rollouts from the initial image o_0 using the same action sequence, and classify an imagined trajectory as a failure if its minimum predicted safety score along the rollout is below zero. Steering is performed by minimizing the learned safety-score predictor for 10 optimization steps.

Baselines. We compare STRESSDREAM against random sampling without optimization, *Nominal* generation with $N = 1$ and *Best-of-N* with $N = 10$, matching the sampling budget of STRESSDREAM. We also compare against STRESSDREAM without the typical-set regularizer $\mathcal{C}^{\text{dist}}$, and against classifier guidance (CG) [56, 57], which applies the safety score gradients during denoising rather than optimizing the initial noise.

STRESSDREAM accurately detects failures only when they are plausible. Figure 2a reports the average minimum safety score over each imagined trajectory, together with the ground-truth minimum safety score of the evaluation trajectories shown as a dotted line.

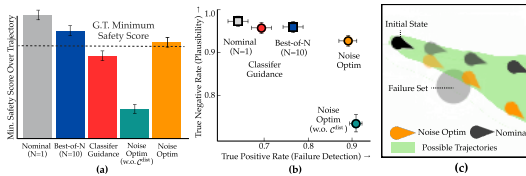


Figure 2: *Results: Naughty Dubins Car.* STRESSDREAM steers WM imaginations plausibly toward failure set, robustly detecting possible failures that random generations miss.

STRESSDREAM achieves accurate pessimistic steering over plausible outcomes, effectively flagging potential failures only when risk exists. In contrast, both noise optimization without the typical-set objective in Sec. 5.2 and CG trivially drive the safety score downward, but do so with a high false-positive rate, indicating implausible imaginations.

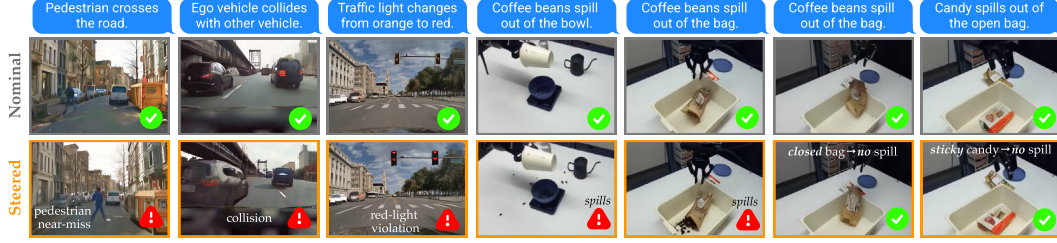


Figure 3: *Nominal vs. Steered Generations*. Top text shows the inference-time prompt describing the target event. STRESSDREAM steers WM imaginations toward critical outcomes that nominal generation misses, such as safety-critical or task-failure events. This steering remains grounded in plausible futures, failing to generate target events when they are not supported by the WM outcome distribution (*right two columns*).

246 7 Robust Policy Evaluation and Improvement with Video World Models

247 We evaluate STRESSDREAM on state-of-the-art video world models for both autonomous driv-
 248 ing [10] and robotic manipulation [11], demonstrating policy evaluation and improvement in real-
 249 world settings.

250 **Autonomous Driving.** We use *Vista* [10] that predicts 25 future front-view frames at 576×1024 ,
 251 conditioned on future waypoints as actions, with noise dimension $D = 921,600$. We fine-tune
 252 the model on the PhysicalAI-Autonomous-Vehicles (PAI-AV) dataset [58] and the Nexar Collision
 253 Prediction Dataset [59] to generate diverse driving scenarios, including collisions. We optimize for
 254 20 steps with Qwen2.5-VL-7B-Instruct [50] fine-tuned with Wolf [60] and X-CLIP [61].

255 **Robotic Manipulation.** We use *Ctrl-World* [11] trained in the DROID setup [62], which predicts
 256 5 future frames from 3 camera views of resolution 192×320 , conditioned on joint-position actions,
 257 with noise dimension $D = 57,600$. We consider six contact-rich tasks (See Fig. 7) and collect
 258 approximately 150 teleoperation trajectories per task containing both successes and failures to fine-
 259 tune the model (More details in Appendix D.1.) We optimize noise for 10 steps with Qwen3-VL-
 260 4B-Instruct [63], providing the three-view videos together with a prompt describing the task failure.

261 7.1 Robust Policy Evaluation via Video World Model Steering by Identifying Risky Actions

262 We first study whether STRESSDREAM enables *robust policy evaluation* by steering WM imagi-
 263 nations, conditioned on given action sequences, toward plausible but critical events specified at
 264 inference time. As a baseline, we mainly compare against random generation (*Best-of-N*), which
 265 samples multiple rollouts from different initial noises and selects the one with the highest score,
 266 without optimization.

267 **Evaluation Setup.** We generate imaginations from an
 268 initial observation o_0 and candidate action sequence $\mathbf{a}$,
 269 and evaluate whether the generated video matches the
 270 inference-time text specification $\mathbf{l}$ of the target criti-
 271 cal event. For driving, we curate 100 image-text pairs
 272 from the PAI-AV dataset [58] across 8 safety-critical
 273 event categories (see Appendix D.3), as well as 200
 274 imminent-collision examples [59]. Each prompt de-
 275 scribes a target event that has not yet occurred in the
 276 initial observation but occurs in the future video. For
 277 manipulation, we collect 200 evaluation trajectories with ground-truth success (100) and failure
 278 (100) labels, and evaluate whether WM imaginations can reliably identify potential failures.

279 **Evaluation Metrics.** For driving, we report *target alignment* and *video quality*
 280 using WorldModelBench [64] (WMB) that is not used for optimization. Target
 281 alignment scores whether the generated video matches the text specification of
 282 target events, while video quality serves as a proxy for plausibility. For manipu-
 283 lation, we label generations as a success or failure based on human judgment.

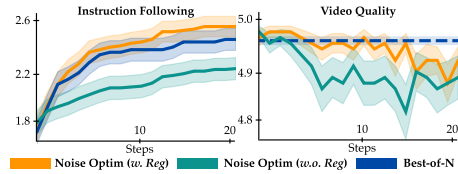


Figure 4: *Steering Driving World Models*. STRESSDREAM steers WM generations toward the target events more effectively than random sampling, while C^{dist} helps preserve video quality.

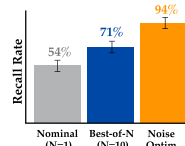


Figure 5: Failure Detection in Manipulation WMs.

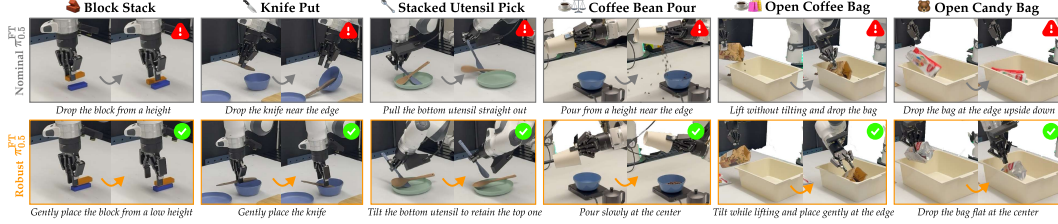


Figure 7: *Robotic Manipulation: Policy Improvement.* Fine-tuning $\pi_{0.5}$ with robust evaluation from steered video world-model imaginations promotes actions that remain successful under pessimistic futures, whereas nominal fine-tuning can reinforce risky actions that happened to succeed during data collection.

284 **STRESSDREAM robustly detects trajectories that risk failures in imagina-**
 285 **tions.** Fig. 3 shows that STRESSDREAM plausibly generates target safety-
 286 critical or failure events often missed by nominal generations, enabling robust
 287 evaluation of action sequences through world-model imaginations. Without the typical-set con-
 288 straint, both target-alignment and video-quality scores decrease, indicating that the distributional
 289 objective helps preserve plausibility (Fig. 4). For manipulation, STRESSDREAM robustly detects
 290 failure events with higher recall, whereas random generations are often overly optimistic and miss
 291 possible failures in imagination (Fig. 5).

292 **STRESSDREAM Steers Only Toward Events Supported by the World Model.** We
 293 study whether steering is grounded in plausible outcomes supported by the WM’s
 294 predictive outcome distribution, rather than hallucinating implausible futures. To
 295 test this, we steer generations toward collision events using both the base *Vista*
 296 model, which is not trained on collision events, and the collision-finetuned model.
 297 Fig. 6 shows that the base model cannot generate collision videos, achieving
 298 even lower target-alignment scores than random sampling from the collision-
 299 finetuned model. This indicates that STRESSDREAM can only steer toward
 300 events that are supported by the world model’s outcome distribution. Similarly, Fig. 3 shows that spilling is induced only when it is plausible under the
 301 world model distribution: STRESSDREAM does not induce spilling for the
 302 open candy bag or closed coffee bag, but does so for the open coffee bag.
 303 Since sticky candies are unlikely to spill and a closed bag does not expose its
 304 contents, spilling is not supported in the WM outcome distribution for those scenes. This shows
 305 that STRESSDREAM discovers possible but plausible outcomes rather than forcing arbitrary desired
 306 events.
 307

308 7.2 Policy Improvement with Steering Video World Models by Promoting Robust Actions

309 **Setup.** We study whether STRESSDREAM can enable improve a behavior-cloning policy, the
 310 Vision-Language-Action (VLA) model $\pi_{0.5}$ [14], through robust action evaluation with WM imag-
 311 inations. We fine-tune $\pi_{0.5}$ -DROID with 40 successful demonstrations per task using a weighted
 312 regression objective [9], and compare two settings: *Nominal* $\pi_{0.5}^{FT}$, which assigns a uniform weight
 313 of 1.0 to all trajectories, and *Robust* $\pi_{0.5}^{FT}$, which assigns weight 1.0 to trajectories that remain suc-
 314 cessful and weight 0.1 to trajectories that fail under steering. This encourages actions that succeed
 315 across the distribution of plausible outcomes induced by the action, while downweighting risky
 316 actions.

317 **STRESSDREAM improves policies by promoting robust actions.** We evaluate
 318 whether STRESSDREAM can improve visuomotor policy learning by promoting rob-
 319 ust actions over risky ones. Fig. 7 illustrates that *Robust* $\pi_{0.5}^{FT}$ favors robust actions
 320 that avoid potential failures, whereas the *Nominal* $\pi_{0.5}^{FT}$ proposes risky actions that
 321 happened to succeed during data collection but fail at inference time, achieving higher
 322 success rates by promoting actions that remain successful even under pessimistic fu-
 323 tures (Fig. 8).

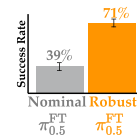


Figure 8: $\pi_{0.5}$ Eval.

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682	Appendix	
683	A Background	20
684	A.1 Extended Related Works	20
685	A.2 Diffusion Models	20
686	A.3 Action-conditioned Video World Models	21
687	A.4 Inference-time Steering of Generative Models	22
688	B Inference-time Steering of Video World Models	23
689	B.1 Regularizing Noise Toward a High-Dimensional Gaussian Prior	23
690	B.2 Concentration of Gaussian Typical Set Regularizers	24
691	B.3 Vision-Language Models as General Inference-Time Steering Specifications	28
692	B.4 Score-Distillation Approximation for Initial-Noise Gradients	30
693	B.5 Implementation Details	31
694	B.6 Time Complexity	32
695	C Naughty Dubins Car	33
696	C.1 Implementation Details	33
697	C.2 Ablation: Optimistic Steering	34
698	C.3 Ablation: Impact of Regularizers for Keeping Generations Plausible	34
699	D Experiment Details	35
700	D.1 Video World Models	35
701	D.2 Robotic Manipulation: Task Details	36
702	D.3 Autonomous Driving: Evaluation Data Curation	37
703	D.4 Evaluation Metrics	37
704	D.5 Fine-tuning $\pi_{0.5}$ -DROID Policy	39
705	E Additional Results	39
706	E.1 Robust Policy Evaluation via Video World Model Steering	39
707	E.2 STRESSDREAM Can Effectively Discover Rare, Long-Tailed Events	42
708	E.3 Video World Model Steering Generates Samples Supported by the Target Distribution	42
709	E.4 Robust Policy Improvement	43
710	F Limitations & Discussions	43

A Background

A.1 Extended Related Works

World Models for Robotics. World models have emerged as a promising paradigm for robotics, enabling the prediction of future states directly from high-dimensional sensory observations conditioned on actions [15, 16, 17, 18, 19, 20, 21, 22, 23]. By generating synthetic trajectories, they have proven effective for vision-based policy learning [65, 66, 67, 44, 68, 69, 70, 9], planning in latent space [19, 71], and policy evaluation [47, 11, 45]. Advances in video generation models [24], which produce realistic and temporally consistent predictions, have further increased the expressiveness of world models in diverse real-world scenarios, especially when fine-tuned with embodiment data for action conditioning [10, 25, 11]. World models typically operate under partial observability, which naturally gives rise to uncertainty and multimodality in future predictions [26, 27, 8], particularly in settings such as contact-rich manipulation [11] or autonomous driving [10, 72, 73, 25]. In particular, diffusion- and flow-matching-based [2, 3, 4] approaches have shown strong performance in modeling high-dimensional, complex environments due to their ability to capture rich, multimodal distributions over future trajectories [5, 6, 7, 8]. Despite this, many existing methods using world models either rely on a single nominal rollout without reasoning over multiple plausible futures [6, 11, 28] or adopt deterministic world models that cannot represent such diversity [19, 23], thereby failing to explicitly account for risks arising from uncertain futures.

Inference-time Steering of Generative Models. Inference-time steering of diffusion [2, 74] or flow-matching [4, 75] model aims to guide a pretrained generative model toward samples that maximize an inference-time reward [13, 29, 30, 31]. While directly fine-tuning diffusion models with reward objectives can be effective for reward alignment [76, 77, 78, 79, 29, 80], it is costly, complex, tied to a specific reward [13, 31], and may shift the model away from the original data distribution [37]. In contrast, inference-time steering keeps the pretrained model fixed and instead modifies the sampling procedure [81, 82, 13, 30, 83, 84], for example through classifier guidance [74] or particle-based sampling methods such as Sequential Monte Carlo (SMC) [85, 86, 87, 31]. One particularly relevant direction is *initial noise optimization*, which optimizes the noise sampled at the start of the denoising process [32, 33, 34, 35, 88, 13]. Prior work has shown that this can improve image quality [33, 36, 37], mimic the effects of diffusion guidance [38], generate rare [12, 39, 40] or diverse concepts [41], and adapt diffusion policies in robotic control [42]. However, a central challenge, especially when the noise space becomes high-dimensional, is out-of-distribution noise: optimized samples may achieve high reward while no longer remaining in the support of the pretrained distribution once the initial noise deviates from the prior [12, 89, 13, 35, 90, 37]. Moreover, these methods often require backpropagating reward gradients through the denoising process; in practice, many therefore rely on step-distilled models to make backpropagation tractable [43, 35, 37, 91], while recent works propose multistep score distillation to approximate reward gradients to the initial noise more efficiently [38]. Our approach adopts initial noise optimization to steer action-conditioned video world models [10, 11] with a practical solution for high-dimensional noise optimization.

A.2 Diffusion Models

Recent generative models are often based on diffusion or flow-matching formulations, which generate samples from noise by simulating a learned differential equation. This section expands on the diffusion-model formulation in Sec. 3, which provides the foundation for recent video world models capable of generating high-quality videos from complex distributions. We omit conditioning on $(\mathbf{o}_{\text{hist}}, \mathbf{a})$ throughout for brevity.

Forward Probability Path. Score-based diffusion models [2, 3, 92] and flow-matching models [93, 4] define generative models f_θ whose sampling process can be described by a stochastic or ordinary differential equation (SDE or ODE). The denoising network g_θ^τ , which parameterizes these

dynamics, is trained to recover the reverse generative process, typically by predicting the score of the perturbed data distribution, the injected noise, or the conditional vector field.

Starting from clean data $\mathbf{x} = \mathbf{o} \sim p_{\text{data}}(\mathbf{o})$ and isotropic Gaussian noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$, the forward process, indexed by diffusion time $\tau \in \{0, 1, \dots, T\}$, defines a perturbed sample as

$$\mathbf{x}^\tau = \alpha^\tau \mathbf{x} + \sigma^\tau \epsilon. \quad (9)$$

For variance-preserving parameterizations such as DDPM, the boundary conditions satisfy $\alpha^0 = 1$, $\sigma^0 = 0$, and $\alpha^T \rightarrow 0$, $\sigma^T \rightarrow 1$. More generally, we only require that the terminal state be noise-dominated. Representative examples include the variance-preserving schedule of DDPM [2], where $(\alpha^\tau)^2 + (\sigma^\tau)^2 = 1$, and the variance-exploding schedule of EDM [3], where $\alpha^\tau \equiv 1$ and σ^τ increases with diffusion time. Our derivation only requires the boundary behavior and smoothness of this schedule.

Training Objective. The denoiser g_θ^τ is trained to recover the injected Gaussian noise from a perturbed future observation, conditioned on the observation history and action sequence. With the noise-prediction parameterization,

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}, \epsilon \sim \mathcal{N}(0, \mathbf{I}), \tau} \left[w(\tau) \left\| \epsilon - g_\theta^\tau(\alpha^\tau \mathbf{x} + \sigma^\tau \epsilon) \right\|_2^2 \right], \quad (10)$$

where $w(\tau)$ is a time-dependent weighting [3]. The clean-sample, score, and velocity parameterizations are linearly equivalent [3, 74]; we take g_θ^τ as a noise predictor, and the analysis transfers to the other parameterizations.

Generation via Deterministic ODE Updates. At inference time, generation starts from an initial Gaussian noise sample $\mathbf{x}^T = \epsilon \sim \mathcal{N}(0, \mathbf{I})$ and iteratively maps it to a clean sample by following the reverse probability-flow ODE. The deterministic update used in (3) can be viewed as a first-order Euler step of this ODE, with DDIM [92] and EDM [3] as representative instances. Given the current noisy sample $\mathbf{x}^\tau$, the predicted clean sample is obtained by inverting the forward path in (9):

$$\hat{\mathbf{x}}^{0|\tau} = \frac{\mathbf{x}^\tau - \sigma^\tau g_\theta^\tau(\mathbf{x}^\tau)}{\alpha^\tau}. \quad (11)$$

The sample is then advanced to the next, less-noisy time $\tau - 1$ by reapplying the forward path to $\hat{\mathbf{x}}^{0|\tau}$, while using the predicted noise as the noise direction:

$$\mathbf{x}^{\tau-1} = \alpha^{\tau-1} \hat{\mathbf{x}}^{0|\tau} + \sigma^{\tau-1} g_\theta^\tau(\mathbf{x}^\tau). \quad (12)$$

Iterating this deterministic update from T to 0 defines the generative model sampling $\mathbf{x} \sim f_\theta$. Substituting (11) into (12) recovers the denoising update in (3).

Probability-Flow ODE and Initial-Noise Determinism. In the continuous-time limit, the deterministic updates in (12) correspond to an Euler discretization of the *probability-flow ODE* associated with the forward path in (9) [74, 3]. This ODE transports samples from the Gaussian prior $\mathcal{N}(0, \mathbf{I})$ to the data distribution p_{data} along deterministic trajectories. Therefore, the generated sample can be written as $\epsilon \mapsto f_\theta(\epsilon) = \mathbf{x}^0$, where f_θ denotes the deterministic sampler induced by the trained denoiser. This makes the generated sample a deterministic and differentiable function of the initial noise, justifying the use of ϵ as a control variable and allowing gradients of downstream criteria to be backpropagated through the sampler.

A.3 Action-conditioned Video World Models

Video world models for robotics have shown promise for policy evaluation and improvement by simulating hard-to-model systems directly from high-dimensional sensor observations [22, 1]. In robotic manipulation, for example, a world model can anticipate how objects may spill, slip, or topple during physical interaction from camera observations [68, 28, 94]. In autonomous driving, it can simulate future scene evolution in complex environments, capturing agent interactions and safety-critical situations directly in image space [10, 25]. These capabilities can support the evaluation and improvement of visuomotor control policies in simulation, reducing reliance on costly, time-consuming, and potentially unsafe real-world trials [1, 46].

Stable Video Diffusion. Recent video world models often build on video generation models [22, 95], which have demonstrated strong ability to synthesize realistic and temporally coherent videos. Rather than learning from scratch, many recent approaches initialize from large-scale video diffusion models and then fine-tune them on domain-specific datasets such as driving or robotic manipulation. A widely used backbone is Stable Video Diffusion (SVD) [24], a latent diffusion model for image-to-video generation. SVD uses an encoder-decoder architecture to map RGB frames into a learned latent space, $z = \mathcal{E}(o) \in \mathbb{R}^{4 \times H/8 \times W/8}$, where H and W denote the image height and width. The diffusion model is then trained to denoise latent video representations, following an objective of the form in (10), and the generated latents are decoded back into RGB video frames using the decoder.

Action Conditioning. For robotic applications, video world models are conditioned on actions to enable controllable future generation [10, 11]. These actions may correspond to future ego-vehicle controls or trajectories in autonomous driving [10, 25], or to robot action sequences such as end-effector poses, joint-position commands, or low-level control commands in robotic manipulation [11, 96]. In the notation of (10), the denoiser is conditioned on histories and actions, $g_\theta^T(\mathbf{x}^T, \mathbf{o}_{\text{hist}}, \mathbf{a})$, so that the generated future is constrained to be consistent with both the observed history and the specified action sequence. While earlier world models often learned latent dynamics from task-specific observation-action trajectories [65, 67, 47], recent diffusion-based video world models built on SVD typically start from a pretrained video diffusion model and post-train it with action-conditioned data. When trained with conditional dropout, these models can also use classifier-free guidance at sampling time to trade off visual realism and action controllability [97].

History Conditioning. To generate coherent future frames that remain consistent with the conditioning observations, recent video world models incorporate observation history in several ways. Some methods inject conditioning frames by replacing the corresponding noisy latent frames with clean observed latents during sampling [10]; others use progressive or autoregressive sampling with per-frame noise schedules to encourage causal structure across time [6, 25]; and some provide sparse history frames or memory tokens as additional network inputs through attention-based conditioning [11]. These mechanisms help preserve temporal continuity, keep generated futures consistent with the observed history, supporting longer-horizon, autoregressive video generation by conditioning future predictions on previous observations or generated frames.

Application: Policy Evaluation & Improvement. Action-conditioned video world models can support robot learning without repeated real-world rollouts, enabling scalable data generation, planning, policy evaluation, and policy improvement. Because current state-of-the-art video generation models are often slow and memory-intensive, they are typically used offline rather than for real-time planning. Prior work has primarily leveraged action-conditioned video prediction in two ways: (i) using imagined rollouts to evaluate or optimize policies without repeated hardware trials [45, 11, 98, 99, 46], and (ii) generating synthetic trajectories for data augmentation or policy fine-tuning [11, 100, 70, 9, 46]. In particular, recent works report that video world models can serve as simulators for closed-loop evaluation of visuomotor policies, with predicted performance correlating with real-world success rates [11]. Similarly, policies can be improved using video world models, either through reinforcement learning inside imagined rollouts [70] or by searching for successful synthetic trajectories that are then used to fine-tune the policy [9].

While our method can be applied to these broader settings, our experiments focus specifically on the value of steering video world models for robust control. Rather than evaluating a complete robotic policy in closed loop, we evaluate candidate trajectories that could be produced by a policy and assess their potential failure risks. Similarly, rather than improving policies through closed-loop exploration inside the world model, we use the steered world model to identify failure-prone expert trajectories and reweight them during policy fine-tuning.

A.4 Inference-time Steering of Generative Models

Given a generative model f_θ that maps an initial noise sample $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ to a generated sample $\mathbf{x} = f_\theta(\epsilon)$, inference-time optimization aims to find generations that maximize a user-specified

851 criterion $\mathcal{C}$, while preserving the plausibility of the generated samples with a regularizer $\text{Reg}(\mathbf{x})$
 852 that penalizes generations that deviate from the original model distribution. This can be written as

$$\mathbf{x}^* = \underset{\mathbf{x}}{\operatorname{argmax}} \mathcal{C}(\mathbf{x}) - \alpha \text{Reg}(\mathbf{x}). \quad (13)$$

853 A natural way to realize this objective is to adapt the generative model itself so that it samples from
 854 a reward-tilted distribution,

$$f_{\theta}^*(\mathbf{x}) \propto f_{\theta}(\mathbf{x}) \exp\left(\frac{\mathcal{C}(\mathbf{x})}{\alpha}\right), \quad (14)$$

855 which upweights high-criterion samples while remaining close to the original model distribution.
 856 However, directly fine-tuning the generator is often impractical for inference-time control: the crite-
 857 rion may change across tasks or users, and fine-tuning requires significant computation for each new
 858 criterion. Moreover, preserving closeness to the original distribution can require density estimates
 859 that are intractable or prohibitively expensive for high-dimensional generative models [77, 101]. As
 860 a result, model adaptation can drift away from the original data distribution and produce implausible
 861 generations [37, 31].

862 **Initial Noise Optimization.** Initial noise optimization instead keeps the generative model fixed and
 863 optimizes only the input noise according to an inference-time criterion evaluated on the generated
 864 sample:

$$\epsilon^* = \underset{\epsilon}{\operatorname{argmax}} \mathcal{C}(f_{\theta}(\epsilon)). \quad (15)$$

865 This formulation allows arbitrary criteria to be specified at inference time without modifying the
 866 model parameters. If the optimized noise remains plausible under the original Gaussian prior, the
 867 resulting generation is still induced by the pretrained model distribution, so additional regularization
 868 on the generated sample itself is not required.

869 When $\mathcal{C}$ is differentiable, the optimization can be solved by backpropagating through the diffu-
 870 sion sampler [32, 34, 35]. For non-differentiable criteria, prior work estimates gradients of the
 871 criterion [13], amortizes the optimization with a hypernetwork [37], or formulates steering as a re-
 872 inforcement learning problem [36, 42]. Since backpropagating through many iterative denoising
 873 steps is computationally expensive, several methods further distill diffusion models into one-step
 874 generators to make optimization more tractable [35, 43, 37].

875 B Inference-time Steering of Video World Models

876 B.1 Regularizing Noise Toward a High-Dimensional Gaussian Prior

877 In this section, we elaborate on Sec. 5.2, where we impose typical-set constraints during noise opti-
 878 mization. In high-dimensional Gaussian distributions, regions of high probability *density* do not
 879 necessarily coincide with regions of high probability *mass*, where typical samples concentrate. For
 880 example, the zero vector has the highest density under a standard Gaussian, yet it is almost never
 881 sampled in high dimensions; instead, typical samples lie on a thin shell with squared norm approx-
 882 imately d . We refer to this high-mass region as the *typical set*. As illustrated in Fig. 9, the typical
 883 set is distinct from the high-density region, and sampling noise outside the typical set can lead to
 884 implausible generations. This distinction is important for diffusion models, which are trained to de-
 885 noise samples initialized from typical Gaussian noise. If optimization pushes the initial noise toward
 886 atypical, low-mass regions of the prior, the denoiser may be evaluated on inputs rarely seen during
 887 training, leading to approximation errors and, consequently, implausible or out-of-distribution gen-
 888 erations.

889 A common way to encourage typicality is to regularize the norm of the optimized noise, since
 890 Gaussian samples exhibit strong norm concentration in high dimensions. Several prior methods have
 891 found norm-based regularization sufficient for preserving plausibility in diffusion generation [12,
 892 35, 13]. However, the norm captures only one statistic of a Gaussian sample. In practice, we find
 893 that additional typicality constraints are important for high-dimensional video world models. Thus,

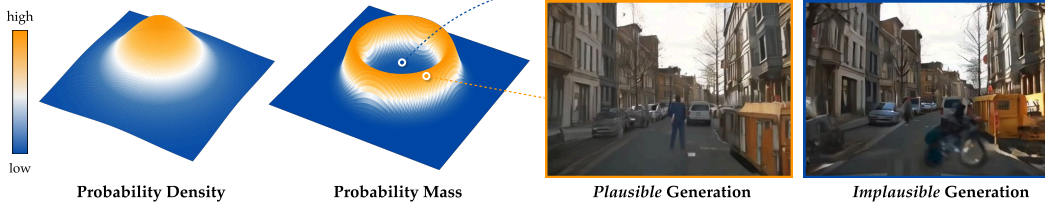


Figure 9: *Probability density vs. probability mass in a high-dimensional Gaussian distribution.* Noise sampled from the typical set leads to plausible generations, whereas noise outside the typical set—despite possibly having high probability density—can produce implausible generations, e.g., humans becoming blurry or transforming into vehicles.

as described in Sec. 5.2, we regularize multiple statistics of the optimized noise, including norm concentration, isotropy, and spectral whiteness, each motivated by concentration properties of high-dimensional Gaussian samples.

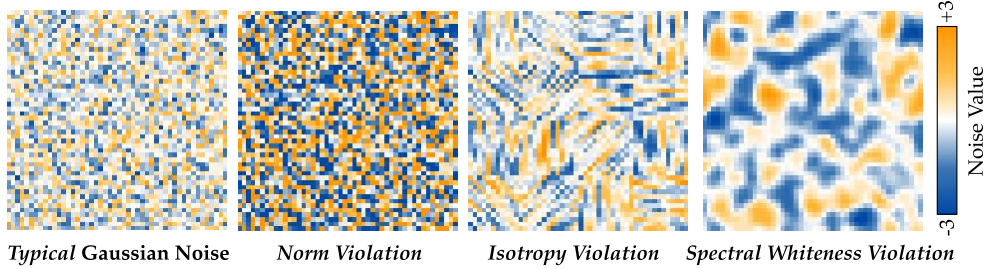


Figure 10: *Examples of atypical Gaussian noise.* Left: typical noise sampled from a Gaussian prior. Others: noises perturbed to violate norm concentration, isotropy, and spectral whiteness, respectively.

Fig. 10 illustrates examples of Gaussian noise visualized as 2D images, where each atypical sample violates one statistic of the Gaussian typical set. Violating norm concentration produces noise with globally biased values compared to typical Gaussian noise. Violating isotropy introduces local coordinate patterns that do not appear in typical i.i.d. noise, while violating spectral whiteness produces structured, non-white patterns in the frequency domain.

B.2 Concentration of Gaussian Typical Set Regularizers

Norm Concentration. For Gaussian noise $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_D)$, the squared norm satisfies $\|\epsilon\|_2^2 \sim \chi_D^2$, and therefore concentrates sharply around D . Equivalently, the Euclidean norm, or radius $\|\epsilon\|_2$ concentrates around $\sqrt{D}$. Moreover, if $R = \|\epsilon\|_2$, then R follows a chi distribution with D degrees of freedom, whose standard deviation remains bounded independently of D and approaches $1/\sqrt{2}$ as $D \rightarrow \infty$. Thus, although the typical radius grows as $\sqrt{D}$, the absolute width of the shell containing most Gaussian probability mass remains approximately constant. This motivates penalizing deviations from the typical Gaussian shell in radius:

$$\mathcal{C}_{\text{norm}}^{\text{dist}}(\epsilon) := \left(\|\epsilon\|_2 - \sqrt{D} \right)^2.$$

Lemma 1 (Norm concentration of Gaussian noise). *Let $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_D)$. Then, for any $\delta \in (0, 1)$,*

$$\mathbb{P} \left(\left| \frac{\|\epsilon\|_2^2}{D} - 1 \right| > 2\sqrt{\frac{\log(2/\delta)}{D}} + 2\frac{\log(2/\delta)}{D} \right) \leq \delta.$$

Equivalently, the probability that ϵ lies outside this typical norm shell is at most δ .

912 *Proof.* Since $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_D)$, its coordinates $\epsilon_1, \dots, \epsilon_D$ are independent standard normal random
 913 variables. Hence,

$$\|\epsilon\|_2^2 = \sum_{j=1}^D \epsilon_j^2 \sim \chi_D^2.$$

914 Let $X = \|\epsilon\|_2^2$. By the Laurent–Massart inequality [102], for any $x > 0$,

$$\mathbb{P}\left(X - D \geq 2\sqrt{Dx} + 2x\right) \leq e^{-x}, \quad \mathbb{P}\left(D - X \geq 2\sqrt{Dx}\right) \leq e^{-x}.$$

915 Taking $x = \log(2/\delta)$, each tail event has probability at most $\delta/2$. Therefore, by a union bound, with
 916 probability at least $1 - \delta$,

$$-2\sqrt{D \log(2/\delta)} \leq X - D \leq 2\sqrt{D \log(2/\delta)} + 2\log(2/\delta).$$

917 Dividing by D gives

$$\left|\frac{X}{D} - 1\right| \leq 2\sqrt{\frac{\log(2/\delta)}{D}} + 2\frac{\log(2/\delta)}{D}.$$

918 Substituting $X = \|\epsilon\|_2^2$ completes the proof. $\square$

919 **Corollary 1** (High-probability scale of the radial norm regularizer). *For*

$$\mathcal{C}_{\text{norm}}^{\text{dist}}(\epsilon) := \left(\|\epsilon\|_2 - \sqrt{D}\right)^2,$$

920 *we have, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\mathcal{C}_{\text{norm}}^{\text{dist}}(\epsilon) \leq \left(2\sqrt{\log(2/\delta)} + 2\frac{\log(2/\delta)}{\sqrt{D}}\right)^2.$$

921 *Consequently,*

$$\mathcal{C}_{\text{norm}}^{\text{dist}}(\epsilon) = O\left(\log(1/\delta) + \frac{\log^2(1/\delta)}{D}\right)$$

922 *with high probability.*

923 *Proof.* By Lemma 1, with probability at least $1 - \delta$,

$$\left|\|\epsilon\|_2^2 - D\right| \leq 2\sqrt{D \log(2/\delta)} + 2\log(2/\delta).$$

924 Let $R = \|\epsilon\|_2$. Since

$$|R - \sqrt{D}| = \frac{|R^2 - D|}{R + \sqrt{D}} \leq \frac{|R^2 - D|}{\sqrt{D}},$$

925 we obtain

$$|R - \sqrt{D}| \leq 2\sqrt{\log(2/\delta)} + 2\frac{\log(2/\delta)}{\sqrt{D}}.$$

926 Squaring both sides gives the desired bound. $\square$

927 **Corollary 2** (Dimension-independent width of the Gaussian shell). *Let $R = \|\epsilon\|_2$ with*
 928 *$\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_D)$. Then the standard deviation of R is bounded independently of D :*

$$\text{std}(R) = O(1).$$

929 *More specifically, R follows a chi distribution with D degrees of freedom, and*

$$\text{std}(R) \rightarrow \frac{1}{\sqrt{2}} \quad \text{as} \quad D \rightarrow \infty.$$

930 *Proof.* Since $R = \|\epsilon\|_2$ and $\|\epsilon\|_2^2 \sim \chi_D^2$, the random variable R follows a chi distribution with D
 931 degrees of freedom. The chi distribution has moments

$$\mathbb{E}[R^2] = D, \quad \mathbb{E}[R] = \sqrt{2} \frac{\Gamma((D+1)/2)}{\Gamma(D/2)}.$$

932 Therefore,

$$\text{Var}(R) = D - 2 \left(\frac{\Gamma((D+1)/2)}{\Gamma(D/2)} \right)^2.$$

933 This variance remains bounded as D grows, and in fact converges to $1/2$. Hence

$$\text{std}(R) \rightarrow \frac{1}{\sqrt{2}},$$

934 showing that the width of the Gaussian shell is asymptotically independent of the dimension. $\square$

935 **Isotropy Concentration.** We now show that the isotropy regularizer is small with high probability
 936 under the Gaussian prior. Let $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_D)$, and suppose $D = mk$. We randomly permute the
 937 coordinates of ϵ and partition them into m blocks

$$\epsilon_1, \dots, \epsilon_m \in \mathbb{R}^k.$$

938 Since the Gaussian prior has i.i.d. coordinates, this random permutation does not change the distri-
 939 bution, and the blocks satisfy

$$\epsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I}_k).$$

940 Thus, their empirical second moment

$$\hat{\Sigma} := \frac{1}{m} \sum_{i=1}^m \epsilon_i \epsilon_i^\top$$

941 should be close to $\mathbf{I}_k$ under the Gaussian prior. We measure deviations from this blockwise isotropy
 942 using

$$\mathcal{C}_{\text{iso}}^{\text{dist}}(\epsilon) := \frac{1}{k} \left\| \hat{\Sigma} - \mathbf{I}_k \right\|_F^2.$$

943 **Lemma 2** (Small probability of anisotropic Gaussian block covariance). *There exists a universal*
 944 *constant $C > 0$ such that, for any $\delta \in (0, 1)$, if $m \gtrsim \log(2k^2/\delta)$, then*

$$\mathbb{P} \left(\mathcal{C}_{\text{iso}}^{\text{dist}}(\epsilon) > C \frac{k \log(2k^2/\delta)}{m} \right) \leq \delta.$$

945 *Proof.* For each entry of $\hat{\Sigma}$,

$$\hat{\Sigma}_{ab} = \frac{1}{m} \sum_{i=1}^m \epsilon_{i,a} \epsilon_{i,b}.$$

946 First, consider the diagonal terms. Since $\epsilon_{i,a} \sim \mathcal{N}(0, 1)$,

$$\hat{\Sigma}_{aa} = \frac{1}{m} \sum_{i=1}^m \epsilon_{i,a}^2, \quad \text{therefore,} \quad \mathbb{E}[\hat{\Sigma}_{aa}] = 1, \quad \text{Var}(\hat{\Sigma}_{aa}) = \frac{2}{m}, \quad \Rightarrow \mathbb{E}[(\hat{\Sigma}_{aa} - 1)^2] = \frac{2}{m}.$$

947 For the off-diagonal terms $a \neq b$, the products $\epsilon_{i,a} \epsilon_{i,b}$ have mean zero and variance one. Hence

$$\mathbb{E}[\hat{\Sigma}_{ab}] = 0, \quad \text{Var}(\hat{\Sigma}_{ab}) = \frac{1}{m}, \quad \text{which gives} \quad \mathbb{E}[\hat{\Sigma}_{ab}^2] = \frac{1}{m}.$$

948 Therefore,

$$\begin{aligned} \mathbb{E} \left[\left\| \hat{\Sigma} - \mathbf{I}_k \right\|_F^2 \right] &= \sum_{a=1}^k \mathbb{E}[(\hat{\Sigma}_{aa} - 1)^2] + \sum_{a \neq b} \mathbb{E}[\hat{\Sigma}_{ab}^2] \\ &= k \cdot \frac{2}{m} + k(k-1) \cdot \frac{1}{m} \\ &= \frac{k^2 + k}{m}. \end{aligned}$$

949 Dividing by k gives

$$\mathbb{E} [C_{\text{iso}}^{\text{dist}}(\epsilon)] = \frac{k+1}{m}.$$

950 For the high-probability statement, note that each diagonal error $\widehat{\Sigma}_{aa} - 1$ is an average of centered
 951 sub-exponential random variables, and each off-diagonal entry $\widehat{\Sigma}_{ab}$ is an average of products of
 952 independent Gaussians, which are also sub-exponential. Therefore, by Bernstein-type concentra-
 953 tion [103], for all entries,

$$\mathbb{P} \left(|\widehat{\Sigma}_{ab} - \delta_{ab}| > t \right) \leq 2 \exp(-cmt^2)$$

954 for $0 < t \lesssim 1$, where $c > 0$ is a universal constant. Applying a union bound over the k^2 entries and
 955 taking

$$t = C' \sqrt{\frac{\log(2k^2/\delta)}{m}},$$

956 we obtain, with probability at least $1 - \delta$,

$$|\widehat{\Sigma}_{ab} - \delta_{ab}| \leq C' \sqrt{\frac{\log(2k^2/\delta)}{m}} \quad \text{for all } a, b.$$

957 Hence,

$$\left\| \widehat{\Sigma} - \mathbf{I}_k \right\|_F^2 \leq k^2 C'^2 \frac{\log(2k^2/\delta)}{m}.$$

958 Dividing by k yields

$$C_{\text{iso}}^{\text{dist}}(\epsilon) \leq C \frac{k \log(2k^2/\delta)}{m}$$

959 with probability at least $1 - \delta$, for a universal constant $C > 0$. □

960 This shows that the normalized isotropy penalty is small under the training-time Gaussian prior
 961 whenever $m \gg k$. Unlike norm concentration, which only constrains the total energy $\text{tr}(\widehat{\Sigma})$, the
 962 isotropy penalty also discourages unequal coordinate variances and nonzero off-diagonal correla-
 963 tions. Therefore, it detects structured coordinate patterns that may still have a typical global norm
 964 but are unlikely under an i.i.d. Gaussian prior.

965 **Spectral-Whiteness Concentration.** The isotropy regularizer checks whether randomly grouped
 966 coordinates have approximately identity covariance, but it does not directly rule out spatial fre-
 967 quency structure. For example, optimized noise may have excessive low-frequency energy, pro-
 968 ducing smooth spatial patterns, or excessive high-frequency energy, producing oscillatory artifacts.
 969 Under an i.i.d. Gaussian prior, the noise should instead be spectrally white: its energy should be
 970 spread evenly across spatial frequencies.

971 To formalize this, consider one spatial slice of the noise and let

$$\tilde{\epsilon} = U\epsilon$$

972 be its discrete Fourier transform, where U is an orthonormal Fourier basis. For real-valued noise, this
 973 can be interpreted as the equivalent real sine-cosine basis. Since the standard Gaussian distribution
 974 is rotationally invariant,

$$\tilde{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

975 Thus, no frequency band should systematically contain more energy than another. Individual Fourier
 976 powers $|\tilde{\epsilon}_j|^2$ are noisy, but averages over sufficiently large frequency bins concentrate.

977 Let $\{\mathcal{B}_b\}_{b=1}^B$ be disjoint frequency bins, and define the average power in each bin by

$$\widehat{p}_b := \frac{1}{|\mathcal{B}_b|} \sum_{j \in \mathcal{B}_b} |\tilde{\epsilon}_j|^2.$$

978 We penalize deviations from a flat binned spectrum using

$$\mathcal{C}_{\text{spec}}^{\text{dist}}(\epsilon) := \frac{1}{B} \sum_{b=1}^B (\widehat{p}_b - \bar{p})^2, \quad \bar{p} := \frac{1}{B} \sum_{b=1}^B \widehat{p}_b.$$

979 **Lemma 3** (Small probability of atypical spectral energy). *Let $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and let $\tilde{\epsilon} = U\epsilon$, where*
 980 *U is an orthonormal discrete Fourier transform. Let $\{\mathcal{B}_b\}_{b=1}^B$ be disjoint frequency bins, and define*
 981 *$\mathcal{C}_{\text{spec}}^{\text{dist}}$ as above. Let*

$$n_{\min} := \min_b |\mathcal{B}_b|.$$

982 *Then, for any $\delta \in (0, 1)$,*

$$\mathbb{P} \left(\mathcal{C}_{\text{spec}}^{\text{dist}}(\epsilon) > 4 \left(2\sqrt{\frac{\log(2B/\delta)}{n_{\min}}} + 2\frac{\log(2B/\delta)}{n_{\min}} \right)^2 \right) \leq \delta.$$

983 *Proof.* Since U is orthonormal and the standard Gaussian distribution is rotationally invariant,

$$\tilde{\epsilon} = U\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

984 Thus, in the equivalent real orthonormal Fourier representation, the Fourier coefficients are indepen-
 985 dent standard normal random variables.

986 For each frequency bin $\mathcal{B}_b$,

$$|\mathcal{B}_b| \hat{p}_b = \sum_{j \in \mathcal{B}_b} |\tilde{\epsilon}_j|^2 \sim \chi_{|\mathcal{B}_b|}^2.$$

987 By the Laurent–Massart inequality [102] and a union bound over the B bins, with probability at
 988 least $1 - \delta$, for all $b = 1, \dots, B$,

$$|\hat{p}_b - 1| \leq 2\sqrt{\frac{\log(2B/\delta)}{|\mathcal{B}_b|}} + 2\frac{\log(2B/\delta)}{|\mathcal{B}_b|}.$$

989 Since $|\mathcal{B}_b| \geq n_{\min}$, we have

$$\max_b |\hat{p}_b - 1| \leq \eta, \quad \eta := 2\sqrt{\frac{\log(2B/\delta)}{n_{\min}}} + 2\frac{\log(2B/\delta)}{n_{\min}}.$$

990 Moreover,

$$|\bar{p} - 1| = \left| \frac{1}{B} \sum_{b=1}^B (\hat{p}_b - 1) \right| \leq \eta.$$

991 Therefore, for every bin,

$$|\hat{p}_b - \bar{p}| \leq |\hat{p}_b - 1| + |\bar{p} - 1| \leq 2\eta.$$

992 Substituting this into the definition of the regularizer gives

$$\mathcal{C}_{\text{spec}}^{\text{dist}}(\epsilon) = \frac{1}{B} \sum_{b=1}^B (\hat{p}_b - \bar{p})^2 \leq 4\eta^2.$$

993 Equivalently,

$$\mathbb{P}(\mathcal{C}_{\text{spec}}^{\text{dist}}(\epsilon) > 4\eta^2) \leq \delta,$$

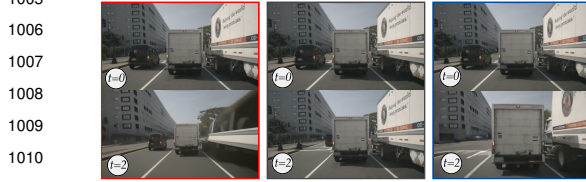
994 which proves the claim. $\square$

995 Lemma 3 shows that the spectral-whiteness penalty is small under the Gaussian prior when each
 996 frequency bin contains sufficiently many coefficients. Thus, a large value indicates an atypical spa-
 997 tial frequency structure. This regularizer is complementary to isotropy: isotropy checks covariance
 998 after random coordinate grouping, whereas spectral whiteness checks whether energy is evenly dis-
 999 tributed across spatial frequencies.

1000 B.3 Vision-Language Models as General Inference-Time Steering Specifications

1001 Rather than learning a task-specific reward function or failure classifier for a narrow set of
 1002 events [68, 28], we use a VLM as an inference-time verifier that can flexibly specify steering tar-
 1003 gets across diverse scenes and event categories [104]. This is particularly useful for general-purpose
 1004 video world models, whose possible futures may vary widely with the scene and action context.

1005 Is the lead vehicle in the ego lane getting *further* from the ego vehicle?



1011 Qwen-VL **0.68** > 0.25 > **0.01**
1012 X-CLIP **0.62** > 0.58 > **0.54**

1013 Figure 11: *VLM score for driving scenes.* The three
1014 videos are generated from the same scene but exhibit
1015 different temporal relationships to the truck in the ego
1016 lane. Both the Qwen-VL-based model and X-CLIP cor-
1017 rectly distinguish these temporal relationships.

VLMs also provide semantically meaningful gradient signals: task-specific reward models trained on limited datasets may fail to cover the large space of possible video generations, making them vulnerable to *reward hacking* [79, 35], where optimization increases the reward through adversarial or visually meaningless changes rather than by producing the intended event. In contrast, because VLMs are trained on large-scale vision-language data, they can provide a more general semantic signal for whether the generated video depicts the specified event [105].

1018 Since our method relies on dense gradient sig-
1019 nals from VLMs, we use open-source models rather than API-based evaluators [106], despite the
1020 strong scene-understanding capabilities of recent proprietary models. We adopt different VLMs for
1021 autonomous driving and robotic manipulation based on the semantic structure of each domain.

1022 **Autonomous Driving.** In autonomous driving, the target events we aim to steer often depend on
1023 temporal relationships among agents, such as pedestrian motion, vehicle movement, or changes in
1024 relative distance. We empirically find that off-the-shelf Qwen models [50] are less reliable at un-
1025 derstanding driving scenes and their temporal structure. We therefore use the Qwen2.5-VL model
1026 fine-tuned on autonomous driving videos by Li et al. [60], which provides improved driving-scene
1027 understanding. We also use X-CLIP [61], which we find effective at capturing temporal relation-
1028 ships. As shown in Fig. 11, for three generated videos where the relative distance to the lead vehicle
1029 is increasing, unchanged, or decreasing, both the fine-tuned Qwen model and X-CLIP [61] assign
1030 scores that correctly distinguish these cases. For CLIP-style models, we define a positive prompt
1031 l_+ and a negative prompt l_- . We embed both prompts and the generated video into the shared
1032 embedding space, and use the difference in cosine similarities as the steering objective:

$$\mathcal{C}_{\psi}^{\text{test}}(\mathbf{o}) = \text{sim}(e_{\text{video}}(\mathbf{o}), e_{\text{text}}(l_+)) - \text{sim}(e_{\text{video}}(\mathbf{o}), e_{\text{text}}(l_-)).$$

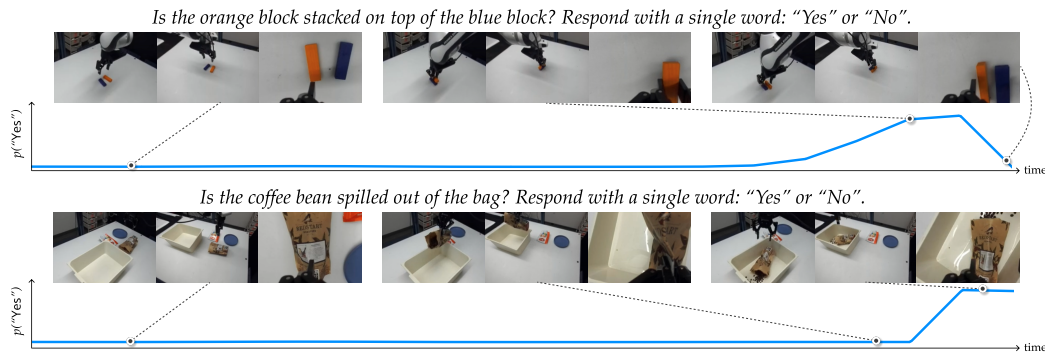


Figure 12: *VLM score for video understanding.* The plot shows the token probability of answering "Yes" to a given prompt. The VLM takes videos from three camera views simultaneously and assigns high scores when the specified events occur, correctly detecting task-relevant moments.

1033 **Robotic Manipulation.** In tabletop manipulation, our steering objective focuses on detecting task-
1034 failure events from generated videos. We use Qwen3-VL [63] with a text prompt that defines the
1035 failure condition, and compute the score from the token probability of the target response, similar
1036 to recent generalizable reward-modeling approaches [107]. Fig. 12 shows that the model score
1037 identifies task-relevant events at the correct moments in generated videos. We provide the five-frame
1038 videos generated by the world model as input and backpropagate dense gradient signals through the
1039 video. We find that using all three camera views simultaneously is essential for reliable scoring:

for the same scene and text prompt, using a single viewpoint leads to substantially poorer failure-event detection. We also find that CLIP-style models, including X-CLIP [61], are less effective at understanding these manipulation scenes.

B.4 Score-Distillation Approximation for Initial-Noise Gradients

This section justifies the gradient approximation used in (7). We omit the conditioning variables $(\mathbf{O}_{\text{hist}}, \mathbf{a})$ for readability. Let the deterministic sampler define the reverse trajectory

$$\mathbf{x}^{\tau-1} = d^\tau(\mathbf{x}^\tau), \quad \mathbf{x}^T = \epsilon, \quad \mathbf{x}^0 = f_\theta(\epsilon),$$

where each denoising step follows the update in (3):

$$d^\tau(\mathbf{x}^\tau) = \frac{\alpha^{\tau-1}}{\alpha^\tau} (\mathbf{x}^\tau - \sigma^\tau g_\theta^\tau(\mathbf{x}^\tau)) + \sigma^{\tau-1} g_\theta^\tau(\mathbf{x}^\tau).$$

For a differentiable criterion $\mathcal{C}(\mathbf{x}^0)$, the exact initial-noise gradient is obtained by the chain rule:

$$\nabla_\epsilon \mathcal{C}(\mathbf{x}^0) = \left[\prod_{\tau=T}^1 \frac{\partial d^\tau}{\partial \mathbf{x}^\tau} \right]^\top \nabla_{\mathbf{x}^0} \mathcal{C}(\mathbf{x}^0),$$

where the product is ordered along the reverse denoising trajectory. The Jacobian of each step is

$$\frac{\partial d^\tau}{\partial \mathbf{x}^\tau} = \frac{\alpha^{\tau-1}}{\alpha^\tau} \mathbf{I} + \left(\sigma^{\tau-1} - \frac{\alpha^{\tau-1}}{\alpha^\tau} \sigma^\tau \right) \mathbf{J}_g^\tau, \text{ where } \mathbf{J}_g^\tau = \frac{\partial g_\theta^\tau(\mathbf{x}^\tau)}{\partial \mathbf{x}^\tau}.$$

Thus, exact backpropagation requires multiplying by all denoiser Jacobians and storing the full denoising computation graph.

Proportional-Jacobian approximation. Ahn et al. [38] empirically observe that the denoiser Jacobian is approximately diagonal and often behaves like a timestep-dependent multiple of the identity. Under the idealized approximation

$$\mathbf{J}_g^\tau \approx \rho^\tau \mathbf{I},$$

the sampler Jacobian becomes

$$\prod_{\tau=T}^1 \left[\frac{\alpha^{\tau-1}}{\alpha^\tau} \mathbf{I} + \left(\sigma^{\tau-1} - \frac{\alpha^{\tau-1}}{\alpha^\tau} \sigma^\tau \right) \mathbf{J}_g^\tau \right] \approx \beta \mathbf{I}, \text{ where } \beta := \prod_{\tau=T}^1 \left[\frac{\alpha^{\tau-1}}{\alpha^\tau} + \left(\sigma^{\tau-1} - \frac{\alpha^{\tau-1}}{\alpha^\tau} \sigma^\tau \right) \rho^\tau \right].$$

Substituting this into the exact chain rule yields

$$\nabla_\epsilon \mathcal{C}(f_\theta(\epsilon)) \approx \beta \nabla_{\mathbf{x}^0} \mathcal{C}(\mathbf{x}^0), \quad \mathbf{x}^0 = f_\theta(\epsilon).$$

This is the approximation used in the main text: instead of differentiating through the sampler, we compute the criterion gradient with respect to the generated clean latent and use it as a scaled proxy for the initial-noise gradient.

Diagonal-Jacobian variant. If the denoiser Jacobian is diagonal but not exactly proportional to the identity,

$$\mathbf{J}_g^\tau \approx \text{diag}(\boldsymbol{\rho}^\tau),$$

then the same derivation gives a diagonal preconditioner

$$\nabla_\epsilon \mathcal{C}(\mathbf{x}^0) \approx \mathbf{B} \nabla_{\mathbf{x}^0} \mathcal{C}(\mathbf{x}^0),$$

where

$$\mathbf{B} = \prod_{\tau=T}^1 \text{diag} \left[\frac{\alpha^{\tau-1}}{\alpha^\tau} \mathbf{1} + \left(\sigma^{\tau-1} - \frac{\alpha^{\tau-1}}{\alpha^\tau} \sigma^\tau \right) \boldsymbol{\rho}^\tau \right].$$

The scalar approximation $\mathbf{B} \approx \beta \mathbf{I}$ is therefore a practical simplification: it assumes the diagonal entries vary slowly enough that their average effect can be absorbed into the optimization step size.

1065 **Connection to stop-gradient MSD.** Multistep score distillation [38] keeps the forward denoising
 1066 trajectory unchanged but applies a stop-gradient operator to the denoiser outputs during backpropa-
 1067 gation. In our notation, the stop-gradient version of one denoising step is

$$d_{\text{SG}}^\tau(\mathbf{x}^\tau) = \frac{\alpha^{\tau-1}}{\alpha^\tau} (\mathbf{x}^\tau - \sigma^\tau \text{SG}[g_\theta^\tau(\mathbf{x}^\tau)]) + \sigma^{\tau-1} \text{SG}[g_\theta^\tau(\mathbf{x}^\tau)],$$

1068 where $\text{SG}[\cdot]$ is the identity in the forward pass and has zero derivative in the backward pass. There-
 1069 fore, this removes $\mathbf{J}_g^\tau$ from the backward pass, leaving only the explicit linear dependence on $\mathbf{x}^\tau$:

$$\frac{\partial d_{\text{SG}}^\tau}{\partial \mathbf{x}^\tau} = \frac{\alpha^{\tau-1}}{\alpha^\tau} \mathbf{I}.$$

1070 Under the same proportional-Jacobian approximation, both the full-gradient and stop-gradient Jaco-
 1071 bian products are scalar multiples of the identity:

$$\prod_{\tau=T}^1 \frac{\partial d^\tau}{\partial \mathbf{x}^\tau} \approx \beta \mathbf{I}, \quad \prod_{\tau=T}^1 \frac{\partial d_{\text{SG}}^\tau}{\partial \mathbf{x}^\tau} = \left(\prod_{\tau=T}^1 \frac{\alpha^{\tau-1}}{\alpha^\tau} \right) \mathbf{I}.$$

1072 Therefore, the stop-gradient gradient and the full sampler gradient differ primarily by timestep-
 1073 dependent scalar factors. Our approximation uses this same principle, but applies it directly to
 1074 optimize the initial noise: the clean-latent criterion gradient is used as a proxy direction for the
 1075 initial-noise gradient, with the unknown scalar absorbed into β and the learning rate.

1076 **Efficacy of Gradient Approximation for Noise Optimization.** The noise-gradient approximation
 1077 enables efficient gradient-based steering without backpropagating through the iterative denoising
 1078 process of large-scale video diffusion models. However, it relies on the assumption that the Jacobian
 1079 of the diffusion sampler is approximately diagonal [38]. Indeed, prior work on noise optimization
 1080 for text-to-image generation often relies on one-step distilled generators to make noise optimization
 1081 tractable without differentiating through the full iterative sampling process.

1082 We therefore empirically evaluate whether this ap-
 1083 proximation provides a useful gradient signal in the
 1084 initial-noise space for generating high-reward sam-
 1085 ples in the output space. Fig. 13 shows the aver-
 1086 age VLM score during noise optimization for steer-
 1087 ing Vista [10] generations in Sec. 7.1. The score in-
 1088 creases throughout optimization, indicating that the
 1089 approximate gradient can improve the output-space
 1090 VLM objective by updating only the initial noise. On an H100 GPU with 80GB VRAM, comput-
 1091 ing the full gradient through the denoising trajectory is impractical even with gradient check-
 1092 pointing [108], whereas the proposed approximation makes the optimization feasible.

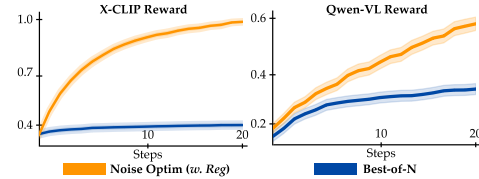


Figure 13:

1093 For Ctrl-World [11], where the full noise gradient can be computed on an
 1094 H100 GPU using gradient checkpointing [108], we compare noise opti-
 1095 mization with and without the gradient approximation. Fig. 14 shows
 1096 that the approximate gradient successfully increases the VLM score,
 1097 whereas optimization with the full gradient fails to improve it. We hy-
 1098 pothesize two possible causes. First, backpropagating through 50 de-
 1099 noising steps may lead to inaccurate or vanishing gradients due to finite-
 1100 precision computation. Second, while the VLM gradient is computed in
 1101 the video space with relatively high numerical precision, the denoising
 1102 process is typically run in lower precision due to memory constraints,
 1103 which may further degrade gradient quality.

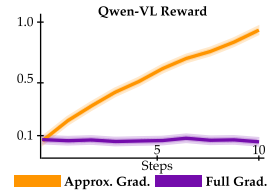


Figure 14: Noise optimization with approximate vs. exact gradients.

1104 B.5 Implementation Details

1105 We summarize the overall procedure of STRESSDREAM in Algorithm 1. To improve optimization
 1106 stability, we use both global gradient-norm clipping and per-coordinate gradient clipping. We addi-
 1107 tionally monitor the Euclidean norm of the noise and project it back to the typical shell, with norm

1108 $\sqrt{D}$, whenever it deviates by more than 3.0. This threshold is conservative: the standard deviation
 1109 of the Gaussian norm is approximately 0.707 in high dimensions, as shown in Corollary 2, so the
 1110 interval contains more than 99.99% of typical Gaussian samples. In addition, we omit momentum
 1111 in the optimizer, since accumulated updates can easily push the noise outside the typical set.

Algorithm 1: STRESSDREAM

Input: Action-conditioned video world model f_θ , Inference-time prompt l , Observation history $\mathbf{o}_{\text{hist}}$, Action sequence $\mathbf{a}$, Optimization steps m , Learning rate η , Regularization weight α , Grad. approx. constant β
Initialize: Initial noise $\epsilon_0 \sim \mathcal{N}(0, \mathbf{I})$;
 Best score $\mathcal{C}^* \leftarrow -\infty$;
 Best generation $\mathbf{o}^* \leftarrow \text{None}$;
 1 **for** $i = 0$ **to** $m - 1$ **do**
 // Generate future observations from current noise)
 2 $\mathbf{o}_i = f_\theta(\epsilon_i, \mathbf{o}_{\text{hist}}, \mathbf{a})$
 // Evaluate Generation with VLMs (Sec. 5.1)
 3 $\mathcal{C}_i = \log p_{\text{VLM}}(\text{yes} | \mathbf{o}_i, l) - \log p_{\text{VLM}}(\text{no} | \mathbf{o}_i, l)$
 // Compute Raw Gradient Components
 4 $\mathbf{g}_{\text{vlm}} = \beta \cdot \nabla_{\mathbf{o}_i} \mathcal{C}_i$;
 5 $\mathbf{g}_{\text{reg}} = \nabla_{\epsilon_i} \mathcal{C}^{\text{dist}}(\epsilon_i)$;
 6 $\mathbf{g}_{\text{total}} = \mathbf{g}_{\text{vlm}} - \alpha \cdot \mathbf{g}_{\text{reg}}$;
 1112 // 1) Gradient clipping for each dimension
 7 $\mathbf{g}_{\text{total}} = \text{clamp}(\mathbf{g}_{\text{total}}, -0.3, 0.3)$;
 // 2) Gradient norm clipping
 8 **if** $\|\mathbf{g}_{\text{total}}\|_2 > 0.5$ **then**
 9 | $\mathbf{g}_{\text{total}} = 0.5 \cdot \frac{\mathbf{g}_{\text{total}}}{\|\mathbf{g}_{\text{total}}\|_2}$;
 // Update Initial Noise via Gradient Ascent
 10 $\epsilon_{i+1} = \epsilon_i + \eta \cdot \mathbf{g}_{\text{total}}$;
 // Hard Normalize to Typical Shell if outside ± 3.0 range
 11 **if** $\|\epsilon_{i+1}\|_2 - \sqrt{D} > 3.0$ **then**
 12 | $\epsilon_{i+1} = \sqrt{D} \cdot \frac{\epsilon_{i+1}}{\|\epsilon_{i+1}\|_2}$;
 13 **if** $\mathcal{C}_i > \mathcal{C}^*$ **then**
 14 | $\mathcal{C}^* = \mathcal{C}_i$;
 15 | $\mathbf{o}^* = \mathbf{o}_i$;
 16 **return** Worst-case generation $\mathbf{o}^*$ and optimized noise ϵ^*

1113 B.6 Time Complexity

1114 We analyze the time complexity of the proposed world model steering algorithm. Let $\mathcal{T}_{\text{denoise}}$ de-
 1115 note the cost of one denoising step and K the number of denoising steps used by the video world
 1116 model. A single deterministic generation costs $\mathcal{T}_{\text{gen}} = K \times \mathcal{T}_{\text{denoise}}$. Let $\mathcal{T}_{\text{verifier}}$ denote the cost of
 1117 evaluating the verifier, such as a VLM-based criterion, on a generated video.

1118 Table 1 compares nominal gener-
 1119 ation, Best-of- N sampling, and
 1120 STRESSDREAM under the same bud-
 1121 get of N sampled candidates or steer-
 1122 ing iterations. Compared to Best-of-
 1123 N sampling, STRESSDREAM adds
 1124 the cost of computing a verifier
 1125 gradient at each steering iteration.
 1126 With the gradient approximation in
 1127 Sec. 5.3, however, STRESSDREAM avoids backpropagating through the full denoising trajectory,
 1128 so the additional cost is mainly the backward pass through the verifier, represented by the second

Method	Time Complexity
Nominal generation	$K \times \mathcal{T}_{\text{denoise}} + \mathcal{T}_{\text{verifier}}$
Best-of- N	$N \times (K \times \mathcal{T}_{\text{denoise}} + \mathcal{T}_{\text{verifier}})$
STRESSDREAM	$N \times (K \times \mathcal{T}_{\text{denoise}} + 2 \mathcal{T}_{\text{verifier}})$
STRESSDREAM w/o Grad. Approx. (Sec.5.3)	$N \times (2K \times \mathcal{T}_{\text{denoise}} + 2 \mathcal{T}_{\text{verifier}})$

Table 1: Time complexity comparison. Here, K is the number of denoising steps, N is the number of sampled candidates or steering iterations, $\mathcal{T}_{\text{denoise}}$ is the cost of one denoising step, and $\mathcal{T}_{\text{verifier}}$ is the cost of one verifier evaluation.

1129 $\mathcal{T}_{\text{verifier}}$ term. Without this approximation, differentiating through the K -step sampler adds a back-
 1130 ward pass through the denoising trajectory, substantially increasing both computation and memory.

1131 In practice, using Vista [10] with $K = 50$ denoising steps on a single H100 GPU, a single gen-
 1132 eration takes approximately 1–2 minutes, while noise optimization with $N = 20$ iterations takes
 1133 about 30 minutes. Since the dominant cost comes from repeated video generation, the runtime of
 1134 STRESSDREAM is expected to improve with faster video world models, such as shortcut [109, 44]
 1135 models.

1136 C Naughty Dubins Car

1137 C.1 Implementation Details

1138 **Video World Model.** We implement a video world model for the Naughty Dubins car environment
 1139 using a smaller-scale SVD-style latent diffusion architecture [24] trained with the EDM formula-
 1140 tion [3]. Following Seo et al. [28], each observation is a rendered $128 \times 128 \times 3$ RGB image
 1141 showing the vehicle and the circular failure set at the center of the environment.

$$\text{Encoder: } z_t = \mathcal{E}_\theta(o_t), \quad \text{Transition: } z_{t+1} \sim f_\theta(z_{t+1} \mid \epsilon, z_t, a_t), \quad \text{Failure: } \hat{\ell}_t = \ell_\theta(z_t).$$

1142 We train the model on an offline dataset of observation-
 1143 action trajectories generated from random initial states
 1144 and random actions, with each trajectory annotated
 1145 by the ground-truth safety margin. Each trajectory
 1146 terminates after $T = 100$ timesteps or earlier if
 1147 the ground-truth x - or y -position leaves the environ-
 1148 ment bounds $[-1.5 \text{ m}, 1.5 \text{ m}]$. The encoder maps ob-
 1149 servations $o_t \in \mathbb{R}^{H \times W \times C}$ to latent representations
 1150 $z_t \in \mathbb{R}^{H/8 \times W/8 \times 4}$, which are decoded back to RGB
 1151 observations using a lightweight decoder. We train the
 1152 latent autoencoder with a reconstruction objective and
 1153 train the failure-margin predictor ℓ_θ using an L_2 regres-
 1154 sion loss against the ground-truth safety margin. We
 1155 then train the latent diffusion transition model f_θ to generate the next latent state conditioned on
 1156 the current latent state and action. In our experiments, we use one-step latent prediction ($H = 1$),
 1157 yielding a latent and noise dimension of 1024. At inference time, we use an Euler sampler with 5
 1158 denoising steps. The world-model hyperparameters are summarized in Tables 2.

WORLD MODEL HYPERPARAMETER	VALUE
IMAGE RESOLUTION	$128 \times 128 \times 3$
LATENT DIMENSION	1024
PREDICTION HORIZON H	1
ACTION DIMENSION	1 (continuous)
VAE BASE CHANNELS	32
VAE LATENT CHANNELS	4
DENOISER HIDDEN DIMS.	[32, 64, 128]
DENOISING STEPS	5
$\sigma_{\min}, \sigma_{\max}$	0.002, 80.0
EDM ρ	7
BATCH SIZE	128
TRAINING EPOCHS	100
LEARNING RATE	1×10^{-4}

Table 2: Dubins Car WM Hyperparameters.

1159 **Evaluation Setup.** We evaluate steering on autoregressive rollouts of the video world model.
 1160 Each rollout starts from the initial image of an evaluation trajectory and is conditioned on the
 1161 corresponding action sequence. We assign a ground-truth label to each evaluation trajectory using
 1162 Monte Carlo rollouts of the true uncertain dynamics in (8). For each initial state and ac-
 1163 tion sequence, we sample 10,000 ground-truth trajectories. We label the trajectory as *posi-*
 1164 *tive* if at least one sampled trajectory enters the failure set, indicating that failure is possible.
 1165 We label it as *negative* if none of the sampled trajec-
 1166 tories enter the failure set. This allows us to eval-
 1167 uate whether pessimistic steering identifies genuinely
 1168 failure-prone trajectories while avoiding spurious fail-
 1169 ures on trajectories that are effectively safe under the
 1170 true dynamics. We steer the generated rollout using the
 1171 hyperparameters in Table 3. Since this setup is small-
 1172 scale, we use the full gradient without the approxima-
 1173 tion. This allows us to directly evaluate the effect of
 1174 optimizing the initial noise and the proposed regularizers
 1175 without introducing additional approximation error from the gradient estimator.

STEERING HYPERPARAMETER	VALUE
OPTIMIZATION ITERATIONS	10
STEP SIZE η IN (7)	1.0
GRADIENT SCALING COEFFICIENT β IN (7)	10.0
NORM CONCENTRATION COEFFICIENT λ_1	1.0
ISOTROPY COEFFICIENT λ_2	0.5
ISOTROPY REGULARIZER SUBVECTOR SIZE k	16
ISOTROPY REGULARIZER PERMUTATION NUMBER	100
SPECTRAL COEFFICIENT λ_3	5.0

Table 3: Dubins Car Steering hyperparameters.

Baseline: Classifier Guidance. We also compare against classifier guidance [56, 57], which modifies the denoising process at each diffusion step using the gradient of the failure criterion. In the world-model setting, however, such guidance is difficult to apply directly. The failure predictor ℓ_θ is trained on clean latent states, not on noisy intermediate diffusion states. Therefore, evaluating ℓ_θ directly on z_t^τ can provide unreliable gradients, while exact guidance would require backpropagating through the denoiser at every denoising step.

To make classifier guidance tractable, we use a one-step clean-latent approximation [57]. Given a noisy latent z_{t+1}^τ , we estimate the corresponding clean latent by inverting the forward diffusion path:

$$\hat{z}_{t+1}^{0|\tau} = \frac{z_{t+1}^\tau - \sigma^\tau g_\theta^\tau(z_{t+1}^\tau, z_t, a_t)}{\alpha^\tau}.$$

The guidance gradient is then computed on this approximately clean latent:

$$\nabla_{z_{t+1}^\tau} \ell_\theta(\hat{z}_{t+1}^{0|\tau}) \approx \frac{1}{\alpha^\tau} \nabla_{\hat{z}_{t+1}^{0|\tau}} \ell_\theta(\hat{z}_{t+1}^{0|\tau}),$$

which avoids backpropagating through the denoising network, but it can introduce bias because the one-step clean prediction may be misaligned with the final generated latent after the full denoising trajectory. This issue is particularly pronounced in action-conditioned world models, where small guidance errors can push the generation toward high-failure latents that are inconsistent with the conditioning action sequence or the learned dynamics.

C.2 Ablation: Optimistic Steering

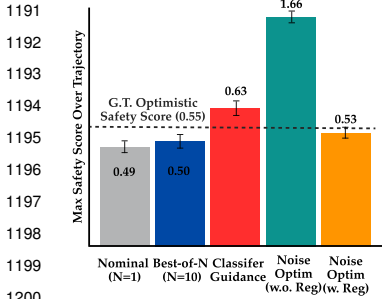


Figure 15: Ablation: *Optimistic Steering of Naughty Dubins Car.*

While Sec. 6 focuses on pessimistic steering to detect potential entries into the failure set, we also ablate the opposite direction by performing optimistic steering. Specifically, we change the sign of the steering objective so that optimization seeks imaginations with higher safety scores. Fig. 15 shows the resulting maximum safety score for each imagined trajectory, together with the maximum safety score achievable under the ground-truth stochastic dynamics as a horizontal reference line. The results are consistent with the findings in Sec. 6: STRESSDREAM can steer the world model effectively, here toward optimistic outcomes, while keeping the generated trajectories plausibly bounded by the ground-truth uncertainty. In contrast, removing the regularizers or using classifier guidance can also increase the imagined safety score, but often produces overly optimistic generations that exceed what is plausible under the ground-truth system dynamics.

C.3 Ablation: Impact of Regularizers for Keeping Generations Plausible

We ablate the impact of the regularizers in Sec. 5.2 on keeping generations plausible, or in-distribution. Unlike video world models trained on internet-scale data, this controlled setting gives us tractable access to the training distribution, allowing us to directly inspect whether steering keeps generations consistent with the training-time distribution. We report typicality scores of the optimized initial noise, including the chi-square log-probability of the squared noise norm, the isotropy regularizer value, and the spectral-whiteness regularizer value. We also remove each regularizer individually to evaluate its contribution.

Empirical Density Estimation. We additionally evaluate whether the generated samples themselves remain *in-distribution* by fitting a flow-matching density surrogate in the latent observation space [110]. Let $z = \mathcal{E}_\theta(o)$ denote the latent representation of an observation from the training dataset. We train a flow-matching vector field q_ψ that transports training latents to standard Gaussian noise. Specifically, for $w \sim \mathcal{N}(0, \mathbf{I})$ and $\tau \in [0, 1]$, define the linear interpolation

$$z^\tau = z + \tau(w - z).$$

1218 The vector field is trained to predict the displacement toward the Gaussian endpoint:

$$q_\psi(z^\tau, \tau) \approx w - z.$$

1219 Given a generated latent $\hat{z}_t$, we estimate its corresponding Gaussian endpoint with a one-step prediction from $\tau = 0$:

$$\hat{w}_t = \hat{z}_t + q_\psi(\hat{z}_t, 0).$$

1221 If $\hat{z}_t$ is in-distribution, then $\hat{w}_t$ should be close to a sample from $\mathcal{N}(0, \mathbf{I})$. Since the Gaussian log-density is proportional to $-\|\hat{w}_t\|_2^2$, we use

$$s_{\text{OOD}}(\hat{z}_t) := \|\hat{w}_t\|_2^2$$

1223 as the latent OOD score. A higher score indicates that the generated latent is farther from the distribution of encoded in-distribution observations. This density-based OOD score complements the noise-space regularizers by checking whether the resulting generated samples, not only the optimized initial noise, remain plausible.

1227 **Results.** Table 4 reports the initial-noise regularizer values and the latent OOD scores of the generated samples. Norm concentration is measured by the minimum chi-square log-probability of the noise norm, where higher values indicate more typical Gaussian noise. Isotropy and spectral whiteness are regularizer values, where lower values are better.

1231 STRESSDREAM with all regularizers keeps the noise statistics close to nominal generation while maintaining a low OOD score. In contrast, steering without regularization produces highly atypical noise and substantially larger OOD scores, suggesting that the optimized generations leave the training distribution. Classifier guidance keeps the initial noise statistics close to nominal because it modifies the denoising trajectory rather than the initial noise, but it still increases the OOD score, indicating that path-level guidance can push generated samples toward implausible regions. Removing individual regularizers also increases the OOD score compared to using all regularizers, highlighting the importance of combining norm concentration, isotropy, and spectral whiteness.

Method	Norm Conc. $\uparrow$	Isotropy $\downarrow$	Spectral White. $\downarrow$	OOD Score $\downarrow$
Nominal	-7.6	0.02	0.03	168.4
Best-of- N	-7.6	0.02	0.03	168.4
Classifier Guidance	-7.6	0.02	0.03	194.5
STRESSDREAM	-6.3	0.02	0.00	171.7
<i>Ablations without regularization</i>				
STRESSDREAM (w/o all reg.)	-717.6	0.83	0.36	696.6
STRESSDREAM (w/o $C_{\text{norm}}^{\text{dist}}$)	-31.7	0.08	0.06	206.1
STRESSDREAM (w/o $C_{\text{iso}}^{\text{dist}}$)	-5.6	0.03	0.11	187.3
STRESSDREAM (w/o $C_{\text{spec}}^{\text{dist}}$)	-6.3	0.00	0.11	172.2

Table 4: Initial-noise regularizer values and latent OOD scores of generated samples. STRESSDREAM with all regularizers keeps the optimized noise close to the Gaussian prior and produces low-OOD generations, whereas removing regularization leads to atypical noise and more out-of-distribution generations.

1248 D Experiment Details

1249 D.1 Video World Models

1250 **Hyperparameters.** Table 5 and Table 6 summarize the video world model and noise-optimization hyperparameters for autonomous driving and robotic manipulation, respectively. For the driving world model, Vista [10]¹, a single generation produces 25 future frames at 10Hz. The action input is an 8-dimensional vector representing the top-down future ego trajectory as four future waypoints. For Ctrl-World [11]², videos are subsampled to 5Hz, and the action input consists of joint-position and gripper commands. Since each prediction generates 5 future frames, the action dimension is 40 = 8 × H. To imagine longer manipulation trajectories, we use autoregressive generation, where the last generated frame is used as the initial conditioning frame for the next prediction window.

1258 **Fine-tuning.** We find that fine-tuning the base checkpoints of the video world models is important for plausibly imagining task-relevant rare events, such as collisions or coffee-bean spills, and for

¹<https://github.com/OpenDriveLab/Vista>

²<https://github.com/Robert-gyj/Ctrl-World>

HYPERPARAMETER	VALUE
NUMBER OF CAMERAS	1
IMAGE RESOLUTION	$576 \times 1024 \times 3$
VIDEO FREQUENCY	10 Hz
PREDICTION HORIZON	25
NOISE DIMENSION	921,600
ACTION DIMENSION	8
DENOISING STEPS	50
CFG SCALE	2.5
OPTIMIZATION ITERATIONS	20
STEP SIZE η	1.0
GRADIENT SCALING β	300.0
NORM COEFFICIENT λ_1	0.5
ISOTROPY COEFFICIENT λ_2	10.0
ISOTROPY SUBVECTOR SIZE k	192
ISOTROPY PERMUTATIONS	1,000
SPECTRAL COEFFICIENT λ_3	100.0

Table 5: World model and steering hyperparameters for Vista [10].

HYPERPARAMETER	VALUE
NUMBER OF CAMERAS	3
IMAGE RESOLUTION	$192 \times 320 \times 3$
VIDEO FREQUENCY	5 Hz
PREDICTION HORIZON	5
NOISE DIMENSION	57,600
ACTION DIMENSION	$40 (8 \times H)$
DENOISING STEPS	50
CFG SCALE	2.0
OPTIMIZATION ITERATIONS	10
STEP SIZE η	1.0
GRADIENT SCALING β	100.0
NORM COEFFICIENT λ_1	0.2
ISOTROPY COEFFICIENT λ_2	0.1
ISOTROPY SUBVECTOR SIZE k	240
ISOTROPY PERMUTATIONS	1,000
SPECTRAL COEFFICIENT λ_3	100.0

Table 6: World model and steering hyperparameters for Ctrl-World [11].

aligning the train- and test-time distributions to improve video quality. While the base models are trained on diverse datasets, such as nuScenes [111] or DROID [62], their generated futures can suffer from degraded dynamics accuracy and visual quality when evaluated on our specific scenarios. We therefore fine-tune each video world model for a small number of iterations on task-relevant data.

Task	Block Stack	Knife Put	Stacked Utensil Pick	Coffee Bean Pour	Open Coffee Bag	Open Candy Bag
# Trajectories	250	170	200	100	170	150

Table 7: Number of teleoperation trajectories used to fine-tune video world model.

For autonomous driving, we fine-tune the video world model using a subset of the PhysicalAI-Autonomous-Vehicles (PAI-AV) dataset [58], nuScenes [111], and the Nexar Collision Prediction Dataset [59]. We train for 10,000 iterations with a batch size of 128. For manipulation, we fine-tune Ctrl-World using teleoperation trajectories that include both successful and failure cases. We use a learning rate of 1×10^{-6} , a batch size of 64, and train for 10,000 iterations. Table 7 summarizes the number of teleoperation trajectories used for each manipulation task.

D.2 Robotic Manipulation: Task Details

We describe the manipulation task setups, the criteria used to determine success or failure, and the language prompts used for steering the generated futures. Visual examples of the task setups are shown in Fig. 7 and on the [project website](#).

Block Stack. Two Jenga blocks, one orange and one blue, are placed on the table. The robot is tasked with stacking the orange block on top of the blue block. A trajectory is considered a *success* if the orange block remains on top of the blue block, and a *failure* if the robot fails to stack the block or if the orange block falls off after stacking. *VLM prompt:* Are the two blocks stacked? Respond with a single word: Yes or No.

Knife Put. A lightweight plastic bowl and a knife are placed on the table. The robot is tasked with picking up the knife and placing it inside the bowl without causing either object to fall. This task is challenging because the knife has an uneven center of mass and a complex shape, while the lightweight bowl can easily tip over. A trajectory is considered a *success* if the knife is placed inside the bowl and remains there, and a *failure* if the knife falls out or the bowl tips over. *VLM prompt:* Is the knife placed inside of the blue bowl without falling out? Respond with a single word: Yes or No.

1286 **Stacked Utensil Pick.** A blue knife and a red spoon are overlapped on a green plate, with the red
1287 spoon placed on top of the blue knife. The robot is instructed to pick up the blue knife while keeping
1288 the red spoon on the plate. A trajectory is considered a *success* if the blue knife is successfully
1289 picked up and the spoon remains on the plate, and a *failure* if the blue knife is not picked up or if the
1290 spoon falls off the plate. The robot should therefore remove the lower utensil while safely leaving
1291 the upper utensil on the plate. *VLM prompt:* Is the red spoon in the green plate? Respond
1292 with a single word: Yes or No.

1293 **Coffee Bean Pour.** A bowl is on top of a coffee scale, and a cup contains coffee beans. The robot
1294 is tasked with pouring the coffee beans from the cup into the bowl on the scale. A trajectory is
1295 considered a *success* if the coffee beans are poured into the bowl without spilling, and a *failure* if
1296 any coffee beans spill onto the table. *VLM prompt:* Is the coffee bean spilled on the table?
1297 Respond with a single word: Yes or No.

1298 **Open Coffee Bag.** An open coffee bag with a variable and partially unobservable amount of coffee
1299 beans inside is on the table, along with a container. The robot is instructed to place the coffee bag
1300 inside the container. A trajectory is considered a *failure* if any coffee beans spill out of the bag, either
1301 into the container or onto the table. Thus, the robot must avoid spilling while lifting, carrying, and
1302 placing the bag into the container. *VLM prompt:* Is the coffee bean spilled? Respond with a
1303 single word: Yes or No.

1304 **Open Candy Bag.** An open gummy bear candy bag is on the table, and the robot is instructed to
1305 place it inside the container without spilling. Compared to coffee beans, gummy bears are stickier
1306 and heavier, creating a smaller risk of spilling out of the bag. A trajectory is considered a *success* if
1307 the gummy bears remain inside the bag, and a *failure* if any gummy bears spill out. *VLM prompt:*
1308 Does candy spill from an open bag? Respond with a single word: Yes or No.

1309 D.3 Autonomous Driving: Evaluation Data Curation

1310 To evaluate steering for autonomous driving video world models, we curate an evaluation set from
1311 the PAI-AV dataset [58]. We focus on scenarios in which the future evolution is uncertain or
1312 interaction-dependent, such as pedestrian crossing, vehicle merging, traffic-light changes, and stop-
1313 sign behavior. Table 8 summarizes the event categories and the verifier prompts used for Qwen-
1314 VL [50] and X-CLIP [61]. For X-CLIP, the first text description is treated as the positive event
1315 description, while the remaining descriptions are used as negative alternatives.

1316 To retrieve candidate clips from the large-scale dataset, we first extract VLM reasoning traces from
1317 Alpamayo-R1 [112] on PAI-AV clips. We then embed these traces using Sentence-BERT [113]
1318 and retrieve the top- k clips whose embeddings are most similar to text descriptions of each event
1319 category. Human annotators then verify whether the target event occurs in each retrieved clip and
1320 label the event occurrence time. Since the video world model generates 2.5s futures, we select the
1321 frame 2.5s before the annotated event time as the initial observation for evaluation. We similarly
1322 curate an evaluation set of 200 collision clips from the Nexar Collision Prediction Dataset [59] by
1323 randomly sampling from the provided collision annotations.

1324 D.4 Evaluation Metrics

1325 **Autonomous Driving.** To evaluate steering, we use held-out metrics that are not used during opti-
1326 mization. This is important because a steered generation can increase the optimization score without
1327 producing a meaningful or plausible change in the video. We therefore evaluate two complementary
1328 aspects: (i) *target alignment*, i.e., whether the generated video clearly depicts the target event de-
1329 scribed by the text prompt, and (ii) *plausibility*, i.e., whether the generated video remains physically
1330 reasonable and free from severe visual artifacts that indicate out-of-distribution generation.

1331 For autonomous driving, we adopt WorldModelBench (WMB) [64], which evaluates video world-
1332 model generations using both text-prompt alignment and video-quality criteria. We use WMB be-

Event Category	Qwen-VL Prompt	X-CLIP Text Descriptions
Pedestrian Crossing	You are a driving assistant interpreting sequential frames from a car's front camera. There is a pedestrian in front of the ego vehicle. Is it true that a pedestrian crosses in front of the ego vehicle? Reply with exactly one word: 'Yes' if a pedestrian crosses in front of the ego vehicle, or 'No' if a pedestrian crossing in front of the ego vehicle is not happening.	Positive: A pedestrian in front of the ego vehicle is crossing the road. Negative: A pedestrian in front of the ego vehicle is not crossing the road. Negative: There is no pedestrian in front of the ego vehicle.
Cyclist Crossing Crosswalk	You are an expert driving assistant interpreting sequential frames from a car's front camera. Does a cyclist in front of the ego vehicle cross a crossway? Reply with exactly one word: 'Yes' if a cyclist crosses a crossway, or 'No' if a cyclist crosses a crossway is not happening.	Positive: A cyclist in front of the ego vehicle crosses a crossway. Negative: A cyclist in front of the ego vehicle does not cross a crossway.
Vehicle Merging from Adjacent Lane	You are an expert driving assistant interpreting sequential frames from a car's front camera. Does a vehicle in the [left right] lane merge into the ego vehicle's lane? Reply with exactly one word: 'Yes' if a vehicle in the [left right] lane merges into the ego vehicle's lane, or 'No' if a vehicle in the [left right] lane merges into the ego vehicle's lane is not happening.	Positive: A vehicle in the [left right] lane merges into the ego vehicle's lane. Negative: A vehicle in the [left right] lane does not merge into the ego vehicle's lane. Negative: A vehicle in the [left right] lane does not move.
Opposing Vehicle Intersection	You are an expert driving assistant interpreting sequential frames from a car's front camera. Does an opposing vehicle at the intersection stop? Reply with exactly one word: 'Yes' if an opposing vehicle at the intersection stops, or 'No' if an opposing vehicle at the intersection stops is not happening.	Positive: An opposing vehicle at the intersection stops. Negative: An opposing vehicle at the intersection does not stop. Negative: An opposing vehicle at the intersection keeps moving.
Lead Vehicle Closing Distance	You are an expert driving assistant interpreting sequential frames from a car's front camera. Is the lead vehicle getting closer to the view of the front camera? The leading vehicle is the vehicle ahead in view only on the same lane, not in the other lane. Reply with exactly one word: 'Yes' if the lead vehicle is getting closer to the view of the front camera, or 'No' if the lead vehicle is not getting closer to the view of the front camera.	Positive: Ego vehicle is getting closer to the leading vehicle Negative: The leading vehicle is getting further away from the ego vehicle Negative: The distance between the ego vehicle and the leading vehicle is not changing
Lead Vehicle Pulling Away	You are an expert driving assistant interpreting sequential frames from a car's front camera. Is the lead vehicle getting further from the view of the front camera? The leading vehicle is the vehicle ahead in view only on the same lane, not in the other lane. Reply with exactly one word: 'Yes' if the lead vehicle is getting further from the view of the front camera, or 'No' if the lead vehicle is not getting further from the view of the front camera.	Positive: The leading vehicle is getting further away from the ego vehicle Negative: Ego vehicle is getting closer to the leading vehicle Negative: The distance between the ego vehicle and the leading vehicle is not changing
Traffic Signal Changing	You are an expert driving assistant interpreting sequential frames from a car's front camera. Initially, the [red orange green] traffic light is on. Is it true that, in the driving scene, the traffic signal visible in the front camera is [red orange green]? Reply with exactly one word: 'Yes' if the traffic signal is [red orange green], or 'No' if the traffic signal is not [red orange green].	Positive: The traffic signal is [red orange green]. Negative: The traffic signal is [red orange green]. Negative: The traffic signal is [red orange green]. Negative: The traffic signal is off.
Front Vehicle Stop-Sign Violation	You are an expert driving assistant interpreting sequential frames from a car's front camera. In the driving scene, the ego vehicle is driving on the road, following the front car. Is it true that, in the driving scene, the front car did not stop at the stop sign? Reply with exactly one word: 'Yes' if a front car did not stop at the stop sign, or 'No' if a front car did not stop at the stop sign.	Positive: The vehicle in front does not stop at the stop sign. Negative: The vehicle in front stops at the stop sign.
Ego-Vehicle Collision	You are a driving safety assistant analyzing sequential frames from a dashcam. The clip shows 2.5 seconds of a driving scene recording. Is a collision or crash between the ego vehicle and another vehicle occurring? A collision means the ego vehicle crashes into another vehicle visible in the front. Reply with exactly one word: 'Yes' if the ego vehicle collides with another vehicle, or 'No' if the situation is safe.	Positive: The ego vehicle collides with another vehicle. Negative: The ego vehicle does not collide with another vehicle. Negative: There is no collision between the ego vehicle and another vehicle.

Table 8: Autonomous driving event categories and verifier prompts used for evaluation data. For X-CLIP, the first description is used as the positive event text and the remaining descriptions are used as negative alternatives.

cause it includes driving data in its evaluation distribution and, empirically, provides meaningful scores for whether generated driving videos both match the target prompt and remain visually plausible. WMB reports three scores: instruction following (0–3), physics adherence (0–5), and commonsense consistency (0–2). We use the instruction-following score to measure target alignment, and use physics adherence and commonsense consistency as proxies for generation plausibility.

In addition, we report a target-alignment score from an external VLM judge. This judge is used only for evaluation and is not used during steering. We use the following prompt template with Gemini [106] to obtain a 0–10 alignment score:

```

You are an expert evaluator of autonomous driving world models. Your ONLY task is to
judge how well the video follows the text instruction. Ignore visual quality issues
such as blur, flickering, warping, or low resolution: these are expected artifacts of
world-model generation and must NOT affect your score.
TEXT DESCRIPTION: {prompt}
SCORING RUBRIC (0-10):

```

1347 10: The video is aligned with the text description clearly and completely.
1348 7-9: The video is aligned with the text description but with minor deviations (e.g.,
1349 slightly wrong direction or timing).
1350 4-6: The video is vaguely related to the text description, but the specific action
1351 does not happen.
1352 1-3: The video is completely unrelated to the text description.
1353 0: The video content is completely unrelated to the text description.

1354 **Robotic Manipulation.** We use human evaluation to determine whether each generated trajec-
1355 tory corresponds to task success or failure. We additionally report scores from Robometer [114], a
1356 general-purpose robotic reward model. However, we rely on human evaluation as the primary metric
1357 for a more accurate assessment of generated trajectories, since reward models provide continuous
1358 scores and require prompt engineering to infer task success.

1359 D.5 Fine-tuning $\pi_{0.5}$ -DROID Policy

1360 We fine-tune the $\pi_{0.5}$ -DROID policy³ using 40 expert trajectories per task with a weighted flow-
1361 matching objective [9]. After steering the imagined rollout of each trajectory toward task failure, we
1362 assign a weight of 1.0 to trajectories classified as successful and a weight of 0.1 to trajectories steered
1363 to failure. We use two camera views, consisting of a wrist camera and one third-person camera. The
1364 weights are applied both to the fine-tuning loss and to dataset sampling. We use joint-position actions
1365 for fine-tuning, although the base policy was originally trained with joint-velocity actions, because
1366 our controller [115] does not support joint-velocity control. We fine-tune the pretrained checkpoint
1367 for 10k steps on a single H100 GPU. At inference time, we use an open-loop action horizon of 16.
1368 We found that training a diffusion policy [116, 117] from scratch performs poorly in this setting,
1369 likely due to the contact-rich nature of our tasks and the limited size of the fine-tuning dataset.
1370 In contrast, fine-tuning $\pi_{0.5}$ -DROID produces substantially better behavior and clearly imitates the
1371 expert demonstrations after fine-tuning.

1372 E Additional Results

1373 E.1 Robust Policy Evaluation via Video World Model Steering

1374 In this section, we provide additional results for Sec. 7.1. The evaluation focuses on steering video
1375 world-model generations toward text-defined failure events in each scene, enabling *robust* evaluation
1376 of policy-proposed action sequences through pessimistic world-model imaginations. We measure
1377 both target alignment, i.e., how well the generated videos match the failure prompt, and plausibil-
1378 ity, i.e., whether the generations remain visually and physically reasonable. Full results, including
1379 videos and interactive demos, are available on the [project website](#).

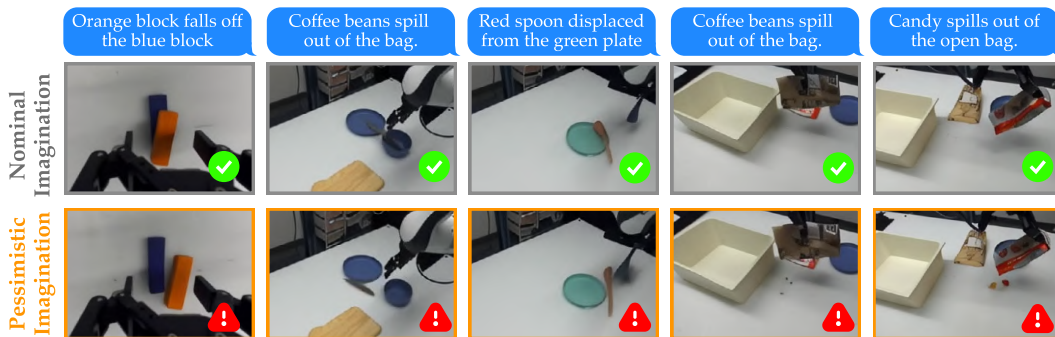


Figure 16: *Additional Results: Nominal Generation vs. Steered Generation.* Given the same observation history and action sequence, STRESSDREAM steers the video world model toward plausible task-failure events when failure is possible, whereas nominal generation often misses these failure modes.

³<https://github.com/Physical-Intelligence/openpi>

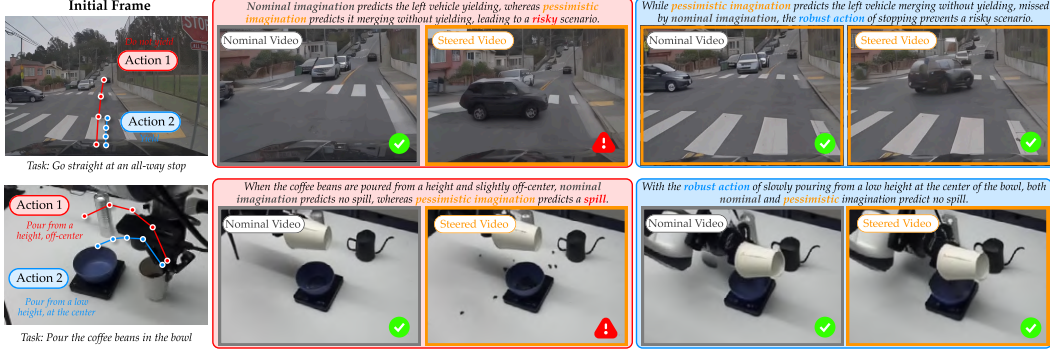


Figure 17: *Robust Policy Evaluation with Video World Models.* Steering the video world model enables robust detection of safety-critical or task-failure events in imagined futures that nominal generation may miss, thereby enabling the selection of *robust* action sequences that remain safe even under worst-case imaginations.

Robust Policy Evaluation with Pessimistic Imaginations. Fig. 16 shows additional results for the inner optimization in (4) on robotic manipulation tasks. For the same observation history and action sequence, nominal imaginations often do not predict failure events, whereas STRESSDREAM generates plausible task-failure futures from the same input. This enables robust policy evaluation using the video world model by detecting possible failures through synthetic trajectories, without requiring repeated hardware rollouts. In contrast, nominal generation may miss such failures because it can sample non-failing futures even when failure is possible. Similarly, Fig. 17 shows that, across different candidate action sequences, STRESSDREAM with pessimistic steering can reveal potential failures or safety-critical events in the imagined futures, while robust action sequences remain safe even under pessimistic imagination. This supports robust evaluation and action selection by using the video world model as a predictive model for planning and control.

Detailed Results: Driving Video World Model. Fig. 18 shows the full quantitative results for driving world-model [11] steering, evaluated with WorldModelBench (WMB) [64] and Gemini-3.0 [106].

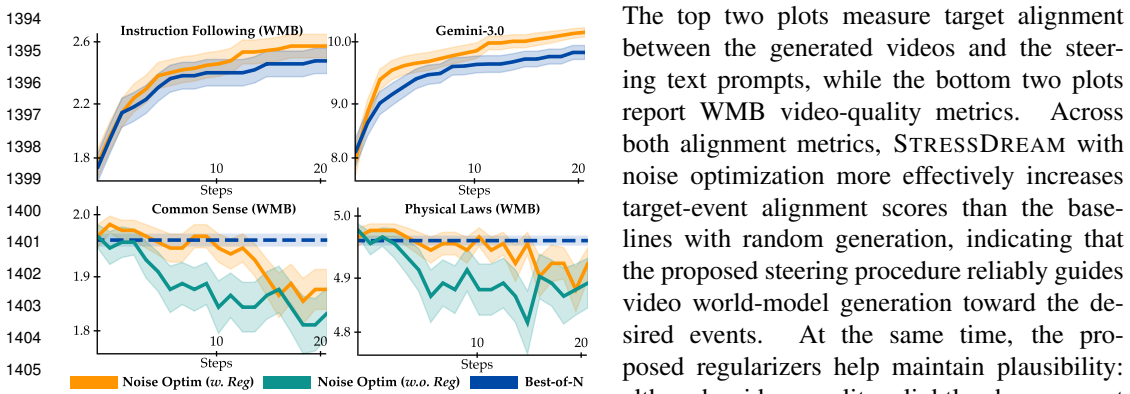


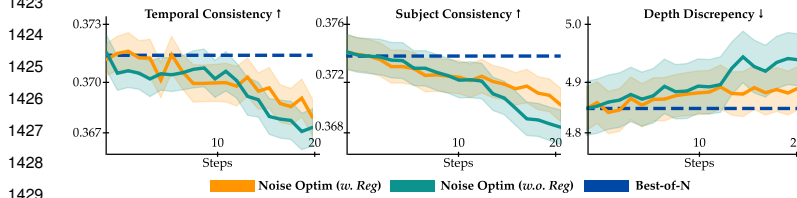
Figure 18: *Quantitative: Driving WM.*

preserves better quality in both common sense scores and physical laws scores than steering without regularization.

Regularizers Preserve Plausibility of Steered Video Generations. We have evaluated the plausibility of steered generations on VLM-based scores, such as the commonsense and physical-law scores in WorldModelBench [64]. To further assess the role of the regularizers in Sec. 5.2, we additionally evaluate generated driving videos using WorldLens [118]. We focus on its generation-quality metrics, which assess whether generated videos are visually realistic, temporally stable, and

1415 semantically consistent without requiring multi-view images or ground-truth labels. Unlike VLM-
 1416 inferred scores, these metrics are computed from analytic feature-based measures⁴.

1417 Specifically, we evaluate *Subject Consistency*, *Depth Discrepancy*, and *Temporal Consistency*. Sub-
 1418 ject Consistency uses DINO features [119] to measure whether dynamic subjects maintain consis-
 1419 tent texture, shape, and structure over time. Depth Discrepancy measures the temporal stability
 1420 of depth representations inferred from generated videos using DepthAnything [120]. Temporal
 1421 Consistency measures frame-to-frame stability in a learned appearance space using CLIP encod-
 1422 ings [121]. We refer readers to the appendix of WorldLens [118] for detailed metric definitions.



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 1429 Figure 19: Video generation quality of driving world-model generations
 1430 evaluated with WorldLens. The regularizers help preserve subject consistency,
 1431 depth stability, and temporal consistency during noise optimization.

1433 proposed noise regularizers substantially mitigates this degradation, preserving subject consistency,
 1434 depth stability, and temporal consistency compared to steering without regularization. These results
 1435 further support that the regularizers are essential for keeping steered generations plausible while
 1436 optimizing for rare or safety-critical events.

1437 **Detailed Results: Failure Detection in Robotic Manipulation.** Fig. 21 reports task-wise failure-
 1438 detection results for robotic manipulation video world models. Steering is performed using Qwen3-
 1439 VL [63] scores computed from task-failure prompts, while the detection results are evaluated against
 1440 trajectory-level success and failure labels from the dataset. Across tasks, STRESSDREAM detects
 1441 substantially more failure-prone trajectories than nominal generation or Best-of- N sampling.

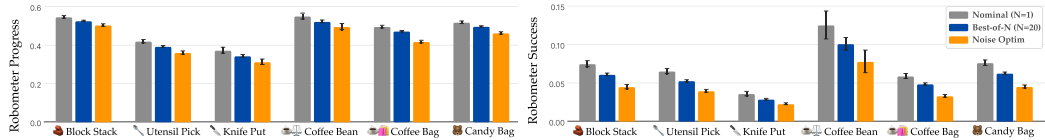


Figure 20: Robometer reward scores for imagined evaluation trajectories. Left: Robometer task-progress
 score. Right: Robometer task-success score. STRESSDREAM effectively generates more pessimistic imagina-
 tions than the baselines, yielding lower task-progress and task-success scores for the same action inputs. This
 indicates that steering can reveal potential failures in the video world model that are missed by nominal gen-
 erations.

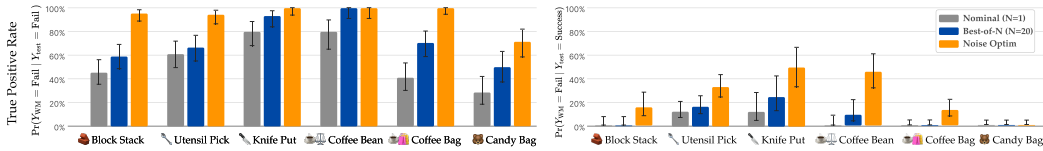


Figure 21: Failure detection in robotic manipulation video world models. Left: recall rate for failures, measur-
 ing the percentage of ground-truth failure trajectories for which the steered imagination predicts failure. Right:
 percentage of imagined failure trajectories among ground-truth non-failure trajectories. This is not necessarily
 a false-positive rate, since some trajectories may be risky but happen to succeed during data collection.

1442 Fig. 21 also shows that STRESSDREAM sometimes predicts failures for trajectories labeled as non-
 1443 failure in the dataset. This should not be interpreted strictly as a false-positive rate: a trajectory that
 1444 succeeded during data collection may still be risky under unobserved factors in a partially observable
 1445 setting. Qualitatively, many of these flagged trajectories correspond to risky behaviors, suggesting

⁴WorldLens also introduces a learned video score trained from large-scale human annotations of world
 realism, physical plausibility, and behavioral safety; however, since this model is not yet released.

that pessimistic imagination provides a more robust way of evaluating policies. At the same time, STRESSDREAM does not flag all non-failure trajectories as failures, indicating that the method can still identify robust trajectories that remain safe even under pessimistic world-model imagination. These robust trajectories can then be prioritized for policy improvement, rather than relying on demonstrations that may have succeeded only by chance.

E.2 STRESSDREAM Can Effectively Discover Rare, Long-Tailed Events

Our main experiments evaluate whether STRESSDREAM can generate videos that satisfy an inference-time specification while preserving plausibility more effectively than random sampling. This is particularly important for video world models, which operate in high-dimensional spaces where targeted generation is difficult: rare events often occupy complex, multimodal, and low-probability regions of the generative distribution, making random sampling highly sample-inefficient. In this section, we ask a complementary question: can STRESSDREAM effectively *discover* rare events in the target distribution by optimizing the initial noise with VLM guidance?

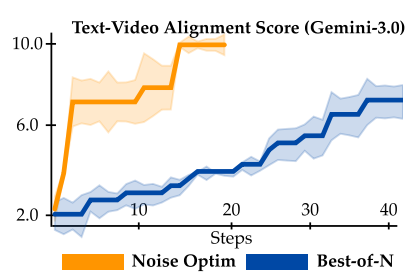


Figure 22: Target-alignment score on long-tailed events in driving video world models.

To evaluate this, we sample 5 evaluation examples from long-tailed driving events in the evaluation set, including pedestrian crossing, traffic-light change, and collision. As a baseline, we generate 40 random samples and report the best result among them, while STRESSDREAM performs 20 optimization steps from a single initial sample. We evaluate all generated videos using Gemini-3.0 [106], which scores how well each video aligns with the target event description.

Fig. 22 shows that STRESSDREAM is substantially more effective at discovering target rare events than random sampling. Even with 40 samples, Best-of- N sampling does not match the target-alignment score achieved by STRESSDREAM. This suggests that first-order optimization in the high-dimensional noise space can discover rare, long-tailed events more efficiently than zeroth-order search through random sampling, which can easily miss such events during generation.

E.3 Video World Model Steering Generates Samples Supported by the Target Distribution

We further elaborate on the experiments in Sec. 7.1 by examining whether steered generations remain grounded in plausible outcomes of the world model. In particular, we compare a driving world model fine-tuned with collision data against the base model without such fine-tuning. As shown in Fig. 6, the base model yields low text–video alignment scores for collision events, even compared to random sampling from the collision-fine-tuned model, because collisions are not represented in its predictive distribution and therefore cannot be plausibly imagined.

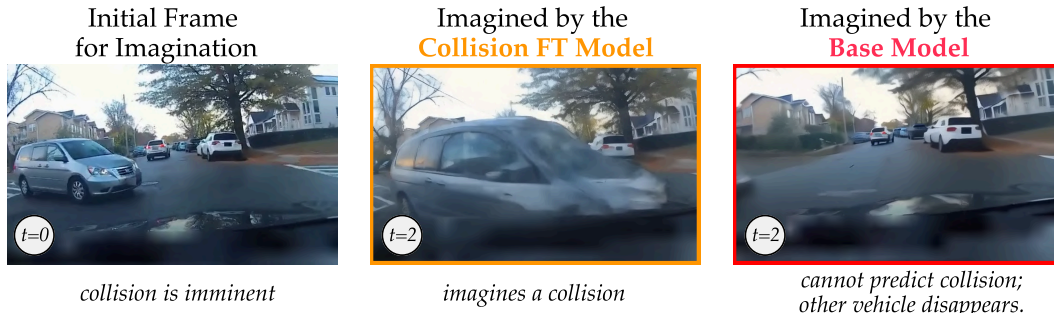


Figure 23: Imaginations near collision from the base model and the collision-fine-tuned world model.

Fig. 23 shows a qualitative example of imagination near a collision event using the base model and the collision-fine-tuned model. The base model fails to imagine the collision and instead hallucinates the other vehicle disappearing from the scene. In contrast, the fine-tuned model, whose predictive

distribution includes collision events, correctly imagines the collision. This supports our claim that when the optimized noise remains close to the Gaussian prior, steering produces samples that are still supported by the world model distribution, rather than arbitrary hallucinations. In this sense, noise optimization differs from simply forcing inference-time alignment through a classifier or a fine-tuning objective, which can more easily push the generation off-manifold.

Similarly, Fig. 24 shows an example of steering a driving-world-model generation so that the distance between the ego vehicle and the lead vehicle decreases. In the initial generation, the lead vehicle moves forward, roughly maintaining its distance from the ego vehicle. In the steered generation, the lead vehicle instead remains stopped while the ego vehicle continues moving, causing the distance to decrease. Although the distance could be reduced even further if the lead vehicle moved backward, such behavior is highly implausible in a normal driving scenario. Accordingly, the steered generation chooses the plausible outcome of the lead vehicle stopping, rather than the implausible outcome of it reversing. By contrast, when steering is performed without the regularizers, the generation leaves the data manifold and hallucinates unrealistic artifacts, such as the distorted backward motion of vehicle shapes, instead of producing a plausible video prediction.

Is the lead vehicle in the ego lane getting *closer* to the ego vehicle?

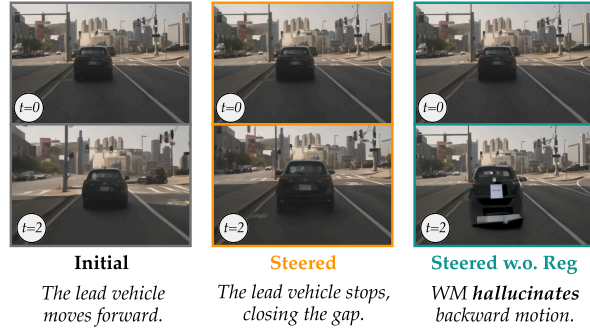


Figure 24: *Qualitative example: steering toward reduced distance to the lead vehicle. With regularization, steering produces a plausible outcome supported by the world model, whereas without regularization, it leads to implausible hallucinations.*

E.4 Robust Policy Improvement

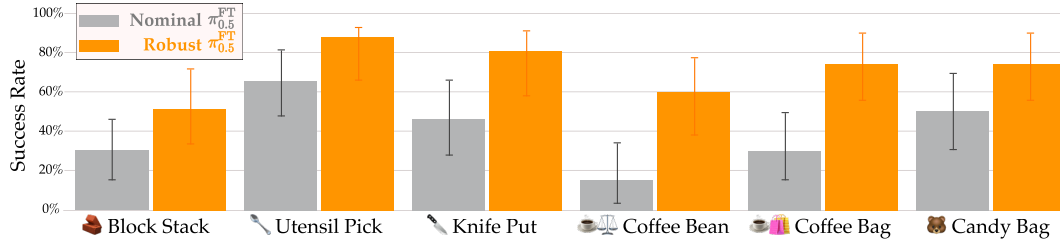


Figure 25: *Task-wise success rates of fine-tuned $\pi_{0.5}$ -DROID policies. Robust fine-tuning with pessimistic video world-model imaginations improves policy success rates across tasks compared to nominal fine-tuning.*

Fig. 25 reports the task-wise success rates of the $\pi_{0.5}$ -DROID policy fine-tuned with video world-model imaginations (Sec. 7.2). Across all tasks, the *Robust* $\pi_{0.5}^{FT}$ policy achieves higher success rates than the *Nominal* $\pi_{0.5}^{FT}$ policy. This suggests that pessimistic imaginations help identify and downweight failure-prone demonstrations, thereby promoting actions that remain successful even under adverse imagined futures. Videos of policy rollouts are available on the [project website](#).

F Limitations & Discussions